Commutative Algebra

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I. A First Course

§1 Valuation Rings

Definition 1.1. An integral domain R with field of fractions K is said to be a *valuation ring* if for every $x \in K^\times$, $x \in R$ or $x^{-1} \in R$. We also say that R is a valuation ring of K.

Proposition 1.2. Let R be a valuation ring of K. The R-submodules of K are totally ordered with respect to inclusion.

Proof. Let I and J be distinct R-submodules of K. If $I \setminus J \neq \emptyset$, pick $x \in I \setminus J$. For any $0 \neq y \in J$, one of xy^{-1} or $x^{-1}y$ must belong to R. If $xy^{-1} \in R$, then $x = y \cdot (xy^{-1}) \in J$, which is absurd. Thus $x^{-1}y \in R$, so that $y = x \cdot (x^{-1}y) \in I$, and hence, $J \subseteq I$. Argue similarly for the case $J \setminus I \neq \emptyset$. ■

Corollary 1.3. A valuation ring is local.

Proposition 1.4. The following are equivalent for a ring R:

(1) R is a valuation ring.

(2) R is a local Bézout domain.

Proof. (1) $\implies$ (2) follows immediately from Proposition 1.2, since any finitely generated ideal in R can be written as $a_1R + \cdots + a_nR$ for some $a_1, \ldots, a_n \in R$.

(2) $\implies$ (1): We must show that for every $x \in K^\times$, either $x \in R$ or $x^{-1} \in R$. Choose $f, g \in R$ such that $x = \frac{f}{g}$, and let $(h) = (f, g)$ as ideals in R. We can find $a, b \in R$ such that $f = ah$ and $g = bh$, and can find $c, d \in R$ such that $h = cf + dg$. Since R is a domain and $h \neq 0$, it follows that $ac + bd = 1$. Therefore, either a or b must be a unit in R, so that either $x \in R$ or $x^{-1} \in R$, thereby completing the proof. ■

Proposition 1.5. A valuation ring is integrally closed.

Proof. Let $(R, \mathfrak{m})$ be a valuation ring with fraction field K. If $x \in K \setminus R$ is integral over R, then there is a non-trivial relation of the form

$$x^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0$$

with $a_i \in R$ for $0 \leqslant i \leqslant n-1$. Multiplying out by $x^{-n} \in \mathfrak{m}$, we have that $1 \in \mathfrak{m}$, a contradiction. Thus R is integrally closed. ■

REMARK 1.6. We note here that given a field K, a valuation ring of K is determined by its maximal ideal. Indeed, if $(R,\mathfrak{m})$ is a valuation ring of K, then we can write

$$K \setminus R = \{x^{-1} : x \in \mathfrak{m} \setminus \{0\}\}.$$

Further note that if R is a valuation ring of K, then any subring of K containing R is also a valuationg ring of K.

THEOREM 1.7. Let R be a valuation ring of K and R' a subring of K containing R. Let $\mathfrak{m}$ denote the maximal ideal of R, $\mathfrak{p}$ the maximal ideal of R', and suppose that $R \neq R'$. Then

(1) $\mathfrak{p} \subsetneq \mathfrak{m} \subseteq R \subseteq R'$.

(2) $\mathfrak{p}$ is a prime ideal of R and $R' = R_{\mathfrak{p}}$.

(3) $R/\mathfrak{p}$ is a valuation ring of the field $R'/\mathfrak{p}$.

(4) Given a valuation ring $\overline{S}$ of the field $R/\mathfrak{m}$, let S denote its preimage in R. Then S is a valuation ring of K and is called the *composite* of R and $\overline{S}$.

Proof. (1) Let $0 \neq x \in \mathfrak{p}$. Then $x^{-1} \in K \setminus R' \subseteq K \setminus R$, so that $x \in \mathfrak{m}$. Thus $\mathfrak{p} \subseteq \mathfrak{m}$. In light of Remark 1.6, $\mathfrak{p} \neq \mathfrak{m}$.

(2) Since $R/\mathfrak{p} \hookrightarrow R'/\mathfrak{p}$ as a subring, it is clear that $\mathfrak{p}$ is a prime ideal in R. Further, since every element of $R \setminus \mathfrak{p}$ is invertible in R', $R \subseteq R_{\mathfrak{p}} \subseteq R'$. Note that $\mathfrak{p}$ is an ideal in $R_{\mathfrak{p}}$, and hence, $\mathfrak{p}R_{\mathfrak{p}} = \mathfrak{p}$, so that $R_{\mathfrak{p}} = R'$ due to (1).

(3) Straightforward.

(4) Let $\pi : R \to R/\mathfrak{m}$ denote the projection map. Note that $\mathfrak{m} \subseteq S$ and $S/\mathfrak{m} = \overline{S}$. If $x \in R \setminus S$, then $\pi(x) \notin \overline{S}$, and therefore, $\pi(x^{-1}) = \pi(x)^{-1} \in S$. Hence $x^{-1} \in S$. On the other hand, if $x \in K \setminus R$, then $x^{-1} \in \mathfrak{m} \subseteq S$. This shows that S is a valuation ring of K. ■

THEOREM 1.8. Let K be a field, $A \subseteq K$ a subring, and $\mathfrak{p}$ a prime ideal of A. Then there exists a valuation ring $(R,\mathfrak{m})$ of K such that $A \subseteq R$, and $\mathfrak{m} \cap A = \mathfrak{p}$.

Proof. First, replacing A by $A_{\mathfrak{p}}$, we may assume that A is a local ring with maximal ideal $\mathfrak{p}$. Let $\mathscr{F}$ denote the set of all subrings B of K containing A such that $1 \notin \mathfrak{p}B$. Clearly, every ascending chain in $\mathscr{F}$ has an upper bound given by the union of all elements of the chain. Using Zorn's lemma, choose a maximal element R in $\mathscr{F}$. First, we contend that R is a local ring. Indeed, since $1 \notin \mathfrak{p}R$, there is a maximal ideal $\mathfrak{m}$ of R containing $\mathfrak{p}R$. Note that $R_{\mathfrak{m}} \in \mathscr{F}$, so that $R = R_{\mathfrak{m}}$ in view of the maximality of R. Next, $\mathfrak{m} \cap A$ is a proper ideal containing $\mathfrak{p}$, which, due to the maximality of $\mathfrak{p}$ must be equal to $\mathfrak{p}$.

It remains to show that R is a valuation ring of K. Suppose $x \in K$ is such that $x, x^{-1} \notin R$. Then $R \subsetneq R[x]$, and hence $1 \in \mathfrak{p}R[x]$, i.e., there is a polynomial relation

$$1 = a_0 + a_1 x + \cdots + a_n x^n$$

with $a_i \in \mathfrak{p}R \subseteq \mathfrak{m}$ for $0 \leqslant i \leqslant n$. Multiplying by $(1-a_0)^{-1} \in R$, one obtains a relation of the form

$$1 = b_1 x + \cdots + b_n x^n$$

with $b_i \in \mathfrak{p}R$ for $1 \leqslant i \leqslant n$. Choose one such relation with the smallest possible value of n. Arguing similarly for x^{-1}, choose a relation

$$1 = c_1 x^{-1} + \cdots + c_m x^{-m}$$

with the smallest possible value of m. If $n \geqslant m$, then multiply the second equation by $b_n x^n$ and subtract from the first to obtain a non-trivial relation of smaller degree than n, a contradiction. On the other hand, if $m > n$, then multiply the first relation by $c_m x^{-m}$ and subtract from the second to obtain a non-trivial relation of smaller degree than m, a contradiction again. Thus $x \in R$ or $x^{-1} \in R$, i.e., R is a valuation ring of K. ■

THEOREM 1.9. Let K be a field, $A \subseteq K$ a subring, and B the integral closure of A in K. Then B is equal to the intersection of all valuation rings of K containing A.

Proof. Let C denote the intersection of all valuation rings of R containing A. In view of Proposition 1.5, B is contained in every such valuation ring, so that $B \subseteq C$. Now let $x \in K$ be non-integral over A and set $y = x^{-1}$. Note that $1 \notin yA[y]$, else x would be integral over A. Let $\mathfrak{p}$ be a maximal ideal of $A[y]$ containing $yA[y]$, and using Theorem 1.8, choose a valuation ring $(R, \mathfrak{m})$ of K containing $A[y]$ such that $\mathfrak{m} \cap A[y] = \mathfrak{p}$. In particular, $y \in \mathfrak{m}$, so that $x \notin R$. Thus $x \notin C$, as desired. ■

§§ Valuations

DEFINITION 1.10. An abelian group Γ together with a total order relation $\leqq$ is said to be *ordered* if for all $x, y \in \Gamma$ with $x \leqq y$, $x + z \leqq y + z$ for all $z \in \Gamma$.

Let K be a field. A *valuation* on K is a map $v : K \to \Gamma \cup \{\infty\}$ satisfying:

(i) $v(x) = \infty$ if and only if $x = 0$,

(ii) $v(xy) = v(x) + v(y)$[1], and

(iii) $v(x + y) \geqslant \min\{v(x), v(y)\}$

for all $x, y \in K$.

Corresponding to a valuation v on K, we can define

$$R_v = \{x \in K : v(x) \geqslant 0\} \quad \text{and} \quad \mathfrak{m}_v = \{x \in K : v(x) > 0\}.$$

It is not hard to see that $(R_v, \mathfrak{m}_v)$ is a valuation ring of K. Conversely, we show that every valuation ring arises this way. Let

$$\Gamma = \left\{xR : x \in K^\times\right\}.$$

This is a group under the operation

$$(xR) \cdot (yR) = xyR$$

with neutral element R. Further, note that Γ is an ordered abelian group when equipped with the total ordering

$$xR \leqq yR \iff xR \supseteq yR,$$

where we are implicitly invoking Proposition 1.2. Define $v : K \to \Gamma \cup \{\infty\}$ sending

$$v(x) = \begin{cases} xR & x \neq 0 \\ \infty & x = 0. \end{cases}$$

Note that $xR \geqq R$ if and only if $xR \subseteq R$, that is, if $x \in R$. Hence, $R_v = R$ and $\mathfrak{m}_v = \mathfrak{m}$. This shows that every valuation ring arises from a valuation of K.

[1] so that v is a group homomorphism when restricted to $K^\times$

Definition 1.11. The *value group* of a valuation ring $(R, \mathfrak{m})$ of K is $v(K^\times)$, where v is a valuation of K such that $(R_v, \mathfrak{m}_v) = (R, \mathfrak{m})$.

To see that the value group is well-defined:

Proposition 1.12. If v and v' are two valuations of K corresponding to the same valuation ring R and having value groups H and H' respectively, then there is an order-isomorphism $\varphi\colon H \to H'$ such that $v' = \varphi \circ v$.

Proof. Let $v\colon K^\times \to \Gamma$ and $v'\colon K^\times \to \Gamma'$ be the two valuations. Define $\varphi\colon H \to H'$ by $\varphi(v(x)) = v'(x)$ for all $x \in K^\times$. We must show that φ is well-defined. Indeed, if $x, y \in K^\times$ are such that $v(x) = v(y)$, then $x^{-1}y$ is a unit in R, so that $v'(x^{-1}y) = 0$, i.e., $v'(x) = v'(y)$. That φ is a group homomorphism is clear. The surjectivity of φ is immediate from the fact that $v' = \varphi \circ v$. As for injectivity, if $\varphi(v(x)) = 0$, then $v'(x) = 0$, and hence x is a unit in R, so that $v(x) = 0$. Finally, if $v(x) \leqq v(y)$, then $v(x^{-1}y) \geqq 0$, so that $x^{-1}y \in R$, i.e., $v'(x^{-1}y) \geqq 0$, that is, $v'(y) \geqq v'(x)$. Thus φ is an order-isomorphism. ■

§2 Discrete Valuation Rings and Dedekind Domains

§§ Discrete Valuation Rings

Definition 2.1. A valuation ring with value group order-isomorphic to $\mathbb{Z}$ is called a *discrete valuation ring (DVR)*.

Theorem 2.2. Let R be a valuation ring. Then the following are equivalent:

(1) R is a DVR.

(2) R is a PID.

(3) R is Noetherian.

Proof. Let K be the field of fractions of R and $\mathfrak{m}$ its maximal ideal.
(1) $\implies$ (2) Let $v\colon K^\times \to \mathbb{Z}$ be a surjective valuation corresponding to R. Let $t \in \mathfrak{m}$ be such that $v(t) = 1$. For $0 \neq x \in \mathfrak{m}$, $v(x) = n > 0$ for some positive integer n. Then $v(x/t^n) = 0$, i.e., $x = ut^n$ for some unit $u \in R^\times$. In particular, this shows that $\mathfrak{m} = tR$. Now let $0 \neq I$ be a proper ideal in R, and let

$$n = \min\{v(a)\colon 0 \neq a \in I\}.$$

Clearly n is a positive integer since I is proper. Let $x \in I$ with $v(x) = n$. Then $xR \subseteq I$, and for every $y \in I$, $v(y/x) \geqslant 0$, so that $y \in xR$. Thus $I = xR$, and R is a PID.
(2) $\implies$ (3) is clear.
(3) $\implies$ (2) A Noetherian Bézout domain is a PID.
(2) $\implies$ (1) Let $t \in \mathfrak{m}$ be such that $\mathfrak{m} = tR$. Recall that a PID is a UFD, and let v denote the t-adic valuation on K. It is not hard to see that R is the valuation ring corresponding to v. ■

Definition 2.3. If R is a DVR with maximal ideal $\mathfrak{m}$, then any element $t \in \mathfrak{m}$ such that $\mathfrak{m} = tR$ is said to be a *uniformizer* or a *uniformizing element* of R.

Remark 2.4. We note here that a valuation ring whose maximal ideal is principal need not necessarily be a DVR. Indeed, let K be a field, $(R, \mathfrak{m}_R)$ a DVR of K, set $k = R/\mathfrak{m}_R$, and suppose $\mathfrak{R}$ is a DVR of k. Let S denote the composite of R and $\mathfrak{R}$. We contend that S is our desired counterexample.

Let $f \in \mathfrak{m}_R$ be a uniformizer of R, $g \in S$ such that $\overline{g} = g + \mathfrak{m}_R$ is a uniformizer of $\mathfrak{R}$. Then $\mathfrak{m}_S = \mathfrak{m}_R + gS$. Further, since $g \notin \mathfrak{m}_R$, it is a unit in R, so that $g^{-1} \in R$. Hence, for any element $h \in \mathfrak{m}_R$, we can write $h = g \cdot \left(g^{-1}h\right)$, so that $\mathfrak{m}_R \subseteq gS$. Hence $\mathfrak{m}_S = gS$ is principal.

Next, we show that S is not Noetherian. Indeed, consider the ascending chain of ideals in S:

$$(f) \subseteq (f, fg^{-1}) \subseteq (f, fg^{-1}, fg^{-2}) \subseteq \cdots.$$

We claim that all the above inclusions are proper. Indeed, if $fg^{-(n+1)} \in (f, fg^{-1}, \ldots, fg^{-n})$ for some $n \geqslant 0$, then there exist $a_0, \ldots a_n \in S$ such that

$$fg^{-(n+1)} = a_0 f + a_1 fg^{-1} + \cdots + a_n fg^{-n}.$$

Multiplying out by $f^{-1}g^{n+1}$, we obtain

$$a_0 g^{n+1} + \cdots + a_n g = 1,$$

which is absurd, since $1 \notin \mathfrak{m}_S = gS$, thereby completing the proof.

THEOREM 2.5. Let R be a ring. The following conditions are equivalent:

(1) R is a DVR.

(2) R is a local PID but not a field.

(3) R is a Noetherian local ring, $\dim R > 0$, and the maximal ideal of R is principal.

(4) R is a one-dimensional normal Noetherian local domain.

Proof. (1) $\Longrightarrow$ (2) $\Longrightarrow$ (3) is clear.

(3) $\Longrightarrow$ (1) Let $\mathfrak{m} = tR$ denote the maximal ideal of R. Due to Krull's intersection theorem,

$$\bigcap_{n=1}^{\infty} \mathfrak{m}^n = (0).$$

Hence, for every $0 \neq x \in R$, there is a non-negative integer n such that $x \in \mathfrak{m}^n \setminus \mathfrak{m}^{n+1}$. Set $v(x) = n$. Now if $x, y \in R \setminus \{0\}$ are such that $v(x) = n$ and $v(y) = m$, then we can find units $u, v \in R^\times$ such that $x = t^n u$ and $y = t^m v$. This shows that $xy = t^{m+n} uv \neq 0$, so that R is an integral domain and $v(xy) = v(x) + v(y)$. Let K denote the fraction field of R, and set

$$v\left(\frac{a}{b}\right) = v(a) - v(b)$$

for all $a, b \in R \setminus \{0\}$. This is clearly well-defined, and defines a valuation on K whose value group is $\mathbb{Z}$ and corresponding valuation ring is R. Hence R is a DVR.

(1) $\Longrightarrow$ (4) Recall that in a DVR the only ideals are (0) and powers of the maximal ideal. Thus the only prime ideals are (0) and $\mathfrak{m}$, so that $\dim R = 1$. That R is Noetherian follows from it being a PID, and finally recall that every valuation ring is normal.

(4) $\Longrightarrow$ (3) Let $\mathfrak{m}$ denote the maximal ideal of R. Due to Nakayama's lemma and the fact that $\dim R = 1$, $\mathfrak{m} \neq \mathfrak{m}^2$. Choose $x \in \mathfrak{m} \setminus \mathfrak{m}^2$. We shall show that $\mathfrak{m} = xR$. Note that $\mathfrak{m} \in \mathrm{Ass}_R(R/xR)$, so that there exists $y \in R \setminus xR$ such that $(xR : y) = \mathfrak{m}$. Set $a = yx^{-1} \in K := \mathrm{Frac}(R)$. Note that $a \notin R$ lest $y \in xR$. Set

$$\mathfrak{m}^{-1} := \{\alpha \in K : \alpha\mathfrak{m} \subseteq R\}.$$

Note that $\mathfrak{m}^{-1}$ is an R-submodule of K, and further, by construction,

$$\mathfrak{m}\mathfrak{m}^{-1} := \{R\text{-submodule of } K \text{ generated by } x_i y_i : x_i \in \mathfrak{m},\ y_i \in \mathfrak{m}^{-1}\} \subseteq R \quad \text{and} \quad R \subseteq \mathfrak{m}^{-1}.$$

In particular, we have the inclusions $\mathfrak{m} \subseteq \mathfrak{m}\mathfrak{m}^{-1} \subseteq R$. Hence, $\mathfrak{m}\mathfrak{m}^{-1} \in \{\mathfrak{m}, R\}$. If $\mathfrak{m}^{-1}\mathfrak{m} = \mathfrak{m}$, then $a\mathfrak{m} \subseteq \mathfrak{m}$. Since $\mathfrak{m}$ is a finite R-module, due to Nakayama's lemma, a must be integral over R, but since R is normal, $a \in R$, a contradiction. Thus $\mathfrak{m}^{-1}\mathfrak{m} = R$. Next, note that $x\mathfrak{m}^{-1} \subseteq R$. If $x\mathfrak{m}^{-1} \subseteq \mathfrak{m}$, then

$$xR = x\mathfrak{m}^{-1}\mathfrak{m} \subseteq \mathfrak{m}^2,$$

a contradiction. Hence, $x\mathfrak{m}^{-1} = R$, so that

$$xR = x\mathfrak{m}^{-1}\mathfrak{m} = \mathfrak{m},$$

thereby completing the proof. ■

§§ Fractional Ideals and Dedekind domains

Definition 2.6. Let R be an integral domain with fraction field K. An R-submodule I of K is said to be a *fractional ideal* if there is a non-zero $\alpha \in R$ such that $\alpha I \subseteq R$.

The standard operations on ideals readily generalize to operations on fractional ideals. Indeed, if I and J are fractional ideals, define

$$\begin{aligned} I + J &= \{x + y : x \in I,\ y \in J\} \\ IJ &= \{R\text{-submodule generated by } x_i y_i : x_i \in I,\ y_i \in J\}. \end{aligned}$$

Clearly, both $I + J$ and IJ are fractional ideals of R. Further, if $S \subseteq R$ is a multiplicative set, then

$$S^{-1}I = \left\{\frac{x}{s} : x \in I,\ s \in S\right\}$$

is a fractional ideal of $S^{-1}R$.

Lemma 2.7. Let R be an integral domain, M and N R-submodules of $K = \mathrm{Frac}(R)$. If N is finitely generated, then

$$S^{-1}(M : N) = \left(S^{-1}M : S^{-1}N\right).$$

Proof. Clearly $S^{-1}(M : N) \subseteq \left(S^{-1}M : S^{-1}N\right)$. Since N is finitely generated, we can write $N = a_1R + \cdots + a_nR$ for some $a_1, \ldots, a_n \in K$. If $x \in \left(S^{-1}M : S^{-1}N\right)$, then for all $1 \leqslant i \leqslant n$, $xa_i \in S^{-1}M$. Therefore, there exist $c_i \in S$ such that $c_i x a_i \in M$, whence $c_i x \in (M : N)$, so that $x \in S^{-1}(M : N)$. ■

Definition 2.8. An R-submodule I of K is said to be *invertible* if there exists an R-submodule J of K such that $IJ = R$.

Proposition 2.9. An invertible R-submodule of K must be a finite R-module.

Proof. Let I be an invertible fractional ideal of R and J an R-submodule of K such that $IJ = R$. Then we can find $a_1, \ldots, a_n \in I$ and $b_1, \ldots, b_n \in J$ such that

$$1 = a_1 b_1 + \cdots + a_n b_n.$$

For each $x \in I$, we can write

$$x = (xb_1)a_1 + \cdots + (xb_n)a_n,$$

and note that $xb_i \in R$ for $1 \leqslant i \leqslant n$. It follows that $I = a_1R + \cdots + a_nR$. ■

COROLLARY 2.10. Every invertible R-submodule of K is a fractional ideal.

REMARK 2.11. Suppose I is an invertible fractional ideal of R, and set

$$I^{-1} := \{\alpha \in K : \alpha I \subseteq R\}.$$

This is clearly an R-submodule of K. Further, note that $II^{-1} \subseteq R$, so that

$$I^{-1} = JII^{-1} \subseteq J.$$

But since $JI \subseteq R$, we clearly have $J \subseteq I^{-1}$. This shows that $J = I^{-1}$.

THEOREM 2.12. Let R be an integral domain and I an R-submodule of K. The following are equivalent:

(1) I is an invertible fractional ideal.

(2) I is a projective R-module.

(3) I is a finite R-module, and for each maximal ideal $\mathfrak{m}$ of R, the fractional ideal $I_{\mathfrak{m}} := IR_{\mathfrak{m}}$ of $R_{\mathfrak{m}}$ is principal.

Proof. (1) $\Longrightarrow$ (2) Let $a_1, \ldots, a_n \in I$ and $b_1, \ldots, b_n \in I^{-1}$ be such that $1 = a_1 b_1 + \cdots + a_n b_n$. As we have seen in the proof of Proposition 2.9, $I = a_1 R + \cdots + a_n R$. Let $\pi : F := \bigoplus_{i=1}^{n} Re_i \to I$ be the map sending $e_i \mapsto a_i$. Define $\sigma : I \to F$ by

$$\sigma(x) = (xb_1)e_1 + \cdots + (xb_n)e_n.$$

It is then clear that $\pi \circ \sigma = \mathbf{id}_I$, so that I is projective.

(2) $\Longrightarrow$ (1) There is a free module $F = \bigoplus_i Re_i$ and an epimorphism $\pi : F \to I$ which splits through an R-linear map $\sigma : I \to F$. Let $a_i = \pi(e_i)$ and $\lambda_i = \pi_i \circ \sigma : I \to R$. Note that every R-linear map $I \to R$ is multiplication by some element of K. Say $\lambda_i(x) = b_i x$ for all $x \in I$. Note that $b_i \in I^{-1}$ for all i. Then

$$\pi(\sigma(x)) = \sum_i a_i b_i x \Longrightarrow \sum_i a_i b_i = 1.$$

Set J denote the R-submodule of K generated by the b_i's. Then $R \subseteq IJ \subseteq R$, and hence I is an invertible fractional ideal.

(1) $\Longrightarrow$ (3) That I is a finite R-module is the content of Proposition 2.9. Further,

$$R_{\mathfrak{m}} = \left(II^{-1}\right)_{\mathfrak{m}} = I_{\mathfrak{m}} \left(I^{-1}\right)_{\mathfrak{m}},$$

so that $I_{\mathfrak{m}}$ is an invertible fractional ideal of $R_{\mathfrak{m}}$. Due to (2), $I_{\mathfrak{m}}$ is a projective $R_{\mathfrak{m}}$-module. But since any two elements of K are $R_{\mathfrak{m}}$-linearly dependent, $I_{\mathfrak{m}}$ must be principal.

(3) $\Longrightarrow$ (1) Note that for every maximal ideal $\mathfrak{m}$ of R,

$$\left(I^{-1}\right)_{\mathfrak{m}} = (R : I)_{\mathfrak{m}} = (R_{\mathfrak{m}} : I_{\mathfrak{m}}) = (I_{\mathfrak{m}})^{-1}.$$

If $II^{-1} \subsetneq R$, then there is a maximal ideal $\mathfrak{m}$ containing II^{-1}, so that

$$\mathfrak{m}R_{\mathfrak{m}} \supseteq \left(II^{-1}\right)_{\mathfrak{m}} = I_{\mathfrak{m}} \left(I^{-1}\right)_{\mathfrak{m}} = I_{\mathfrak{m}} I_{\mathfrak{m}}^{-1} = R_{\mathfrak{m}},$$

a contradiction. This completes the proof. ■

THEOREM 2.13. Let R be a Noetherian domain, and $\mathfrak{p}$ a non-zero prime ideal of R. If $\mathfrak{p}$ is invertible, then $\operatorname{ht}\mathfrak{p} = 1$ and $R_{\mathfrak{p}}$ is a DVR.

Proof. Since $\mathfrak{p}$ is invertible, due to Theorem 2.12, $\mathfrak{p}R_{\mathfrak{p}}$ is a principal ideal. Since $\dim R_{\mathfrak{p}} = \operatorname{ht}\mathfrak{p} > 0$, it follows from Theorem 2.5 that $R_{\mathfrak{p}}$ is a DVR, and hence $\operatorname{ht}\mathfrak{p} = \dim R_{\mathfrak{p}} = 1$. ■

THEOREM 2.14. Let R be a normal Noetherian domain. Then

(1) all prime divisors of a non-zero principal ideal have height 1.

(2) $R = \bigcap_{\operatorname{ht}\mathfrak{p}=1} R_{\mathfrak{p}}$.

Proof. (1) This follows from Krull's Hauptidealsatz.

(2) Suppose $\frac{a}{b} \in \bigcap_{\operatorname{ht}\mathfrak{p}=1} R_{\mathfrak{p}}$ with $b \neq 0$. We shall show that $a \in bR$. If b is a unit in R, then there is nothing to prove. If b is not a unit in R, consider the primary decomposition

$$bR = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_r$$

with corresponding associated primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$. Due to (1), we know that $\operatorname{ht}\mathfrak{p}_i = 1$ for all i, so that there are no embedded associated primes. Localising at $\mathfrak{p}_i$ and contracting back to R, we have

$$\mathfrak{q}_i = bR_{\mathfrak{p}} \cap R \ni a.$$

Therefore,

$$a \in \bigcap_{i=1}^{r} \mathfrak{q}_i = bR,$$

as desired. ■

PORISM 2.15. Let R be a Noetherian domain. If all associated primes of a non-zero principal ideal have height 1, then

$$R = \bigcap_{\operatorname{ht}\mathfrak{p}=1} R_{\mathfrak{p}}.$$

COROLLARY 2.16. Let R be a Noetherian domain. Then R is normal if and only if the following two conditions are satisfied:

(i) for every height 1 prime $\mathfrak{p}$, $R_{\mathfrak{p}}$ is a DVR; and

(ii) all associated primes of a non-zero principal ideal of R have height 1.

Proof. Necessity is the content of Theorem 2.14. For sufficiency, note that due to Porism 2.15, $R = \bigcap_{\operatorname{ht}\mathfrak{p}=1} R_{\mathfrak{p}}$. But since each $R_{\mathfrak{p}}$ is a DVR, it is normal, so that R is normal. ■

DEFINITION 2.17. An integral domain for which every ideal is invertible is called a *Dedekind domain*.

THEOREM 2.18. For an integral domain R, the following conditions are equivalent:

(1) R is a Dedekind domain.

(2) R is either a field or a one-dimensional Noetherian normal domain.

(3) every non-zero ideal of R can be written as a product of a finite number of prime ideals.

Proof. (1) $\implies$ (2) Suppose R is a field. Since every ideal is invertible, it is finitely generated, so that R is Noetherian. If $\mathfrak{p}$ is a non-zero prime ideal of R, then due to Theorem 2.12, $\mathfrak{p}R_{\mathfrak{p}}$ is a principal ideal. Due to Theorem 2.5, $R_{\mathfrak{p}}$ is a DVR and $\operatorname{ht}\mathfrak{p} = 1$. Since R is not a field, every prime ideal is a maximal ideal, and hence, we can write

$$R = \bigcap_{0 \neq \mathfrak{p} \in \operatorname{Spec} R} R_{\mathfrak{p}}.$$

Being the intersection of normal domains, R is also a normal domain.

(2) $\implies$ (1) Let I be an ideal of R. We shall show that I is invertible. For a maximal ideal $\mathfrak{m}$ of R, note that $R_{\mathfrak{m}}$ is a one-dimensional Noetherian normal local ring, which, due to Theorem 2.5, is a DVR. In particular, $IR_{\mathfrak{m}}$ is a principal ideal. Thus, due to Theorem 2.12, I is invertible.

(1) $\implies$ (3) That R is Noetherian follows from (2). We prove (3) by Noetherian induction. Let $\mathscr{F}$ denote the set of all non-zero proper ideals in R that are not products of prime ideals. If $\mathscr{F}$ is non-empty, using the Noetherian-ness of R, choose a maximal elmeent $I \in \mathscr{F}$. Note that every proper ideal of R properly containing I can be expressed as a product of prime ideals. Let $\mathfrak{m}$ be a maximal ideal containing I. Clearly $I \neq \mathfrak{m}$ else it has a trivial expression as a product of prime ideals. Since $R \subseteq \mathfrak{m}^{-1}$, we have $I \subseteq I\mathfrak{m}^{-1} \subseteq \mathfrak{m}\mathfrak{m}^{-1} = R$. If $I\mathfrak{m}^{-1} = I$, then due to Nakayama's lemma, every element of $\mathfrak{m}^{-1}$ would be integral over R, and therefore must be an element of R due to (2), a contradiction. Thus $I\mathfrak{m}^{-1} \supsetneq I$, so that it can be expressed as a product of prime ideals. Multiplying this expression by $\mathfrak{p}$, we see that I can be expressed as a product of prime ideals too, a contradiction. Thus $\mathscr{F}$ is empty, as desired.

(3) $\implies$ (1) We shall show that every prime ideal in R is invertible. The factorization property would then imply the same for all non-zero ideals of R.

Step 1. Suppose I and J are fractional ideals of R such that $B = IJ$ is an invertible fractional ideal. We shall show that I and J are invertible. First, note that

$$I^{-1}J^{-1}B = I^{-1}J^{-1}JI \subseteq R \implies I^{-1}J^{-1} = I^{-1}J^{-1}BB^{-1} \subseteq B^{-1}.$$

Also, since $B^{-1}IJ = R$, we have the two obvious inclusions $B^{-1}I \subseteq J^{-1}$ and $B^{-1}J \subseteq I^{-1}$. Multiplying these two inclusions, we obtain

$$B^{-1} \subseteq I^{-1}J^{-1} \implies I^{-1}J^{-1} = B^{-1}.$$

Finally, note that

$$R = BB^{-1} = (II^{-1})(JJ^{-1}),$$

so that $II^{-1} = JJ^{-1} = R$, i.e., I and J are invertible.

Step 2. Let $\mathfrak{p}$ be a non-zero prime ideal in R. We shall show that for any ideal I of R properly containing $\mathfrak{p}$, $\mathfrak{p} = I\mathfrak{p}$. To this end, it suffices to show that $\mathfrak{p} \subseteq I\mathfrak{p}$. Let $a \in I \setminus \mathfrak{p}$ and set $J = aR + \mathfrak{p}$. It suffices to show that $\mathfrak{p} \subseteq J\mathfrak{p}$ so that we may replace I by J and continue our analysis. Consider prime decompositions of the two ideals

$$I^2 = \mathfrak{p}_1 \cdots \mathfrak{p}_r \quad \text{and} \quad a^2R + \mathfrak{p} = \mathfrak{q}_1 \cdots \mathfrak{q}_s.$$

Clearly each of the $\mathfrak{q}_i$'s contain $\mathfrak{p}$. Since $I^2 \subseteq \mathfrak{p}_i$, we have $I \subseteq \mathfrak{p}_i$, so that $\mathfrak{p} \subseteq \mathfrak{p}_i$. Let $\overline{R} := R/\mathfrak{p}$, and for each $x \in R$, let $\overline{x}$ denote its image in $\overline{R}$. Note that $\overline{R}$ is an integral domain with the factorization property (3), and in $\overline{R}$, we have the decompositions

$$\overline{\mathfrak{p}}_1 \cdots \overline{\mathfrak{p}}_r = \overline{a}^2\overline{R} = \overline{\mathfrak{q}}_1 \cdots \overline{\mathfrak{q}}_s.$$

Since $\overline{a}^2\overline{R}$ is an invertible ideal of $\overline{R}$, due to Step 1, all the $\overline{\mathfrak{p}}_i$'s and the $\overline{\mathfrak{q}}_j$'s are invertible ideals of $\overline{R}$. Now let $\overline{\mathfrak{p}}_1$ be a minimal element of the set $\{\overline{\mathfrak{p}}_1, \ldots, \overline{\mathfrak{p}}_r\}$. Since $\overline{\mathfrak{p}}_1 \supseteq \overline{\mathfrak{q}}_1 \cdots \overline{\mathfrak{q}}_s$, using Prime Avoidance, $\overline{\mathfrak{p}}_1$

must contain one of the $\overline{\mathfrak{q}}_i$'s, say without loss of generality, $\overline{\mathfrak{q}}_1$. An analogous argument would give that $\overline{\mathfrak{q}}_1$ must contain some $\overline{\mathfrak{p}}_j$, which, due to the minimality of $\overline{\mathfrak{p}}_1$ must be equal to $\overline{\mathfrak{p}}_1$ – that is, $\overline{\mathfrak{p}}_1 = \overline{\mathfrak{q}}_1$. Since these ideals are invertible, multiplying with their inverses, we are left with

$$\overline{\mathfrak{p}}_2 \cdots \overline{\mathfrak{p}}_r = \overline{\mathfrak{q}}_2 \cdots \overline{\mathfrak{q}}_s.$$

Continuing in this way, we obtain $r = s$ and $\overline{\mathfrak{p}}_i = \overline{\mathfrak{q}}_i$ for $1 \leqslant i \leqslant r$. In particular, $\mathfrak{p}_i = \mathfrak{q}_i$ for $1 \leqslant i \leqslant r$, so that

$$a^2R + \mathfrak{p} = (aR + \mathfrak{p})^2 = a^2R + a\mathfrak{p} + \mathfrak{p}^2.$$

Hence, every $x \in \mathfrak{p}$ can be written as

$$x = a^2y + az + w \qquad \text{where} \quad y \in R,\ z \in \mathfrak{p},\ \text{and}\ w \in \mathfrak{p}^2.$$

Thus $a^2y \in \mathfrak{p}$ – but since $a \notin \mathfrak{p}$, $y \in \mathfrak{p}$. This gives

$$\mathfrak{p} \subseteq a\mathfrak{p} + \mathfrak{p}^2 = I\mathfrak{p},$$

as desired.
Step 3. Let $0 \neq a \in R$. Then in the factorization $bR = \mathfrak{p}_1 \cdots \mathfrak{p}_r$, all the $\mathfrak{p}_i$'s are maximal. Indeed, if I is any ideal properly containing $\mathfrak{p}_i$, then due to Step 2, $\mathfrak{p}_i = I\mathfrak{p}_i$. As we have already argued, every $\mathfrak{p}_i$ is invertible, so that $I = R$.
Step 4. Let $\mathfrak{p}$ be a non-zero prime ideal in R, and let $0 \neq a \in R$. If $aR = \mathfrak{p}_1 \cdots \mathfrak{p}_r$, then due to Step 3, each $\mathfrak{p}_i$ is maximal. Since $\mathfrak{p} \supseteq aR$, there is an index i such that $\mathfrak{p}_i \subseteq \mathfrak{p}$. The maximality of $\mathfrak{p}_i$ forces $\mathfrak{p}_i = \mathfrak{p}$. In particular, $\mathfrak{p}$ is invertible, thereby completing the proof. ■

§3 Dimension Theory

§§ Hilbert Polynomials

THEOREM 3.1. An $\mathbb{N}$-graded ring $R = \bigoplus_{n \geqslant 0} R_n$ is Noetherian if and only if R_0 is Noetherian and R is a finitely generated R_0-algebra.

Proof. The converse direction follows from Hilbert's basis theorem. We prove the forward direction. Suppose R is a Noetherian ring. Since $R_0 = R/R_+$ where $R_+ = \bigoplus_{n>0} R_n$, it follows that R_0 is Noetherian. Next, since R_+ is a finitely generated ideal, we may pick homogeneous elements $x_1, \ldots, x_r \in R_+$ generating it as an R-module. Using induction on $n \geqslant 0$, we shall show that $R_n \subseteq R_0[x_1, \ldots, x_r]$ for all $n \geqslant 0$. Let $d_i > 0$ denote the degree of x_i. Then

$$R_n = \sum_{i=1}^{r} x_i R_{n-d_i},$$

with the convention that $R_j = 0$ for $j < 0$. Using the inductive hypothesis, every element of R_{n-d_i} can be written as a polynomial in the x_i's with coefficients in R_0. From the above relation, it follows that every element of R_n can be written as a polynomial in the x_i's with coefficients in R_0. ■

PROPOSITION 3.2. Let $R = \bigoplus_{n \geqslant 0} R_n$ be a Noetherian graded ring, and $M = \bigoplus_{n \geqslant 0} M_n$ a finite graded R-module. Then each M_n is a finite R_0-module.

Proof. The case $M = R$ has been dealt with in the proof of Theorem 3.1. We may choose the generators $\omega_1, \ldots, \omega_r$ of M as an R-module to be homogeneous. Let d_i denote the degree of each ω_i. Note that $M_+ := \bigoplus_{n \geqslant 0} M_n$ is a submodule of M, it follows that $M_0 = M/M_+$ is a finite R-module which is annihilated by R_+. Therefore, M_0 is a finite R-module. Next, for $n > 0$, we can write

$$M_n = \sum_{i=1}^{r} R_{n-d_i} \omega_i.$$

Since each R_j is a finite R_0-module, it follows that each M_n is a finite R_0-module. ■

Definition 3.3. Let $R = \bigoplus_{n \geqslant 0} R_n$ be a Noetherian graded ring with R_0 Artinian. For a finite graded R-module $M = \bigoplus_{n \geqslant 0} M_n$, we define its *Hilbert series* to be

$$P(M,t) := \sum_{n=0}^{\infty} \lambda_{R_0}(M_n) t^n \in \mathbb{Z}[\![t]\!].$$

Note that each M_n is a finite R_0-module, and hence, is Artinian, so that it has finite length as an R_0-module.

Theorem 3.4 (Hilbert-Serre). Let $R = \bigoplus_{n \geqslant 0} R_n$ be a Noetherian graded ring with R_0 Artinian, and let $M = \bigoplus_{n \geqslant 0} M_n$ be a finite graded R-module. Suppose $x_1, \ldots, x_r \in R$ are homogeneous elements such that R is generated by $x_1, \ldots, x_r$ as an R_0-algebra. If each x_i has degree $d_i > 0$, then the Hilbert series $P(M,t)$ is a rational function, that is, there is a polynomial $f(t) \in \mathbb{Z}[t]$ such that

$$P(M,t) = \frac{f(t)}{\prod_{i=1}^{r} \left(1 - t^{d_i}\right)} \quad \text{in } \mathbb{Z}[\![t]\!].$$

Proof. We shall prove this statement by induction on $r \geqslant 0$. If $r = 0$, then $R = R_0$, so that M is an Artinian R-module. Since there is a descending chain of R-submodules

$$M \supseteq \bigoplus_{n \geqslant 1} M_n \supseteq \bigoplus_{n \geqslant 2} M_n \supseteq \cdots,$$

we must have that $M_n = 0$ for $n \gg 0$, that is, $P(M,t)$ is a polynomial in t, thereby establishing the theorem in this case.

Suppose now that $r > 0$. For $n \geqslant 0$, let K_n and L_{n+d_r} denote the kernel and cokernel of the map $M_n \xrightarrow{\cdot x_r} M_{n+d_r}$. The short exact sequence

$$0 \to K_n \to M_n \xrightarrow{\cdot x_r} M_{n+d_r} \to L_{n+d_r} \to 0$$

gives us

$$\lambda(K_n) - \lambda(M_n) + \lambda(M_{n+d_r}) - \lambda(L_{n+d_r}) = 0. \tag{1}$$

The above convention is such that $L_n = M_n$ for $n < d_r$. Set $K = \bigoplus_{n \geqslant 0} K_n$ and $L = \bigoplus_{n \geqslant 0} L_n$. Note that K_n is annihilated by x_r, and since $L = M/x_r M$ as a graded R-module, it is also annihilated by x_r. In particular, K and L are graded $\overline{R} := R/x_r R$-modules. Multiplying the above equation by t^{n+d_r} and summing over $n \in \mathbb{Z}$, we obtain:

$$t^{d_r} P(K,t) - t^{d_r} P(M,t) + P(M,t) - P(L,t) = 0.$$

Since $\overline{R}$ is an R_0-algebra generated by $x_1,\dots,x_{r-1}$, the induction hypothesis applies so that there are polynomials $g(t),h(t)\in\mathbb{Z}[t]$ such that

$$P(K,t)=\frac{g(t)}{\prod_{i=1}^{r-1}(1-t^{d_i})}\quad\text{and}\quad P(L,t)=\frac{h(t)}{\prod_{i=1}^{r-1}(1-t^{d_i})},$$

whence the conclusion follows. ■

Definition 3.5. In Theorem 3.4 if $d_1=\cdots=d_r=1$, then there is an integer polynomial $f(t)\in\mathbb{Z}[t]$ with $f(1)\neq 0$ and a $d\geqslant 0$ such that

$$P(M,t)=\frac{f(t)}{(1-t)^d}\in\mathbb{Z}[\![t]\!].$$

This integer d is denoted by $d(M)$.

Corollary 3.6. If $d_1=\cdots=d_r=1$ in Theorem 3.4 and $d=d(M)$, then there is a polynomial $\varphi_M(X)\in\mathbb{Q}[X]$ such that for $n\gg 0$, $\lambda(M_n)=\varphi_M(n)$.

Definition 3.7. The polynomial $\varphi_M(X)\in\mathbb{Q}[X]$ is called the *Hilbert polynomial* of the graded R-module M.

§§ Samuel Functions

Let $(A,\mathfrak{m})$ be a Noetherian local ring, M a finite A-module, and $\mathfrak{q}$ an $\mathfrak{m}$-primary ideal. The maximal ideal defines filtrations on A and M:

$$A\supseteq\mathfrak{q}\supseteq\mathfrak{q}^2\supseteq\cdots\quad\text{and}\quad M\supseteq\mathfrak{q}M\supseteq\mathfrak{q}^2M\supseteq\cdots.$$

Let

$$\operatorname{gr}_{\mathfrak{q}}(A):=\bigoplus_{n\geqslant 0}\mathfrak{q}^n/\mathfrak{q}^{n+1}\quad\text{and}\quad\operatorname{gr}_{\mathfrak{q}}(M):=\bigoplus_{n\geqslant 0}\frac{\mathfrak{q}^nM}{\mathfrak{q}^{n+1}M}$$

denote the corresponding associated graded objects. Note that $\operatorname{gr}_{\mathfrak{q}}(M)$ is a finite graded $\operatorname{gr}_{\mathfrak{q}}(A)$-module. If $\mathfrak{q}$ is generated as an A-module by $x_1,\dots,x_r$, and $\overline{x}_1,\dots,\overline{x}_r\in\mathfrak{q}/\mathfrak{q}^2$ denote the corresponding images, then $\operatorname{gr}_{\mathfrak{q}}(A)$ is generated as an $A/\mathfrak{q}$-algebra by $\overline{x}_1,\dots,\overline{x}_r$, which are homogeneous elements of degree 1. Further, since $A/\mathfrak{q}$ is Artinian, we may talk about the Hilbert series of $\operatorname{gr}_{\mathfrak{q}}(M)$ with respect to $\operatorname{gr}_{\mathfrak{q}}(A)$. Now define

$$\chi_M^{\mathfrak{q}}(n)=\sum_{i=0}^{n}\lambda_{A/\mathfrak{q}}\left(\frac{\mathfrak{q}^iM}{\mathfrak{q}^{i+1}M}\right)=\lambda_{A/\mathfrak{q}}\left(\frac{M}{\mathfrak{q}^{n+1}M}\right).$$

Recall from Corollary 3.6 that $\lambda_{A/q}\left(\frac{\mathfrak{q}^nM}{\mathfrak{q}^{n+1}M}\right)$ is a polynomial of degree $d\left(\operatorname{gr}_{\mathfrak{q}}(M)\right)$ for $n\gg 0$. It follows that $\chi_M^{\mathfrak{q}}(n)$ is a polynomial of degree $d\left(\operatorname{gr}_{\mathfrak{q}}(M)\right)+1$ for $n\gg 0$.

Proposition 3.8. If $\mathfrak{p}$ and $\mathfrak{q}$ are two $\mathfrak{m}$-primary ideals, then $\deg\chi_M^{\mathfrak{p}}=\deg\chi_M^{\mathfrak{q}}$.

Proof. There are integers $a>0$ and $b>0$ such that $\mathfrak{p}^a\subseteq\mathfrak{q}$ and $\mathfrak{q}^b\subseteq\mathfrak{p}$. Thus, $\mathfrak{p}^{a(n+1)}\subseteq\mathfrak{q}^{n+1}$ and $\mathfrak{q}^{b(n+1)}\subseteq\mathfrak{p}^{n+1}$. In particular, this gives the inequalities

$$\chi_M^{\mathfrak{p}}(a(n+1)-1)\geqslant\chi_M^{\mathfrak{q}}(n)\quad\text{and}\quad\chi_M^{\mathfrak{q}}(b(n+1)-1)\geqslant\chi_M^{\mathfrak{p}}(n)$$

for every positive integer n. Thus $\deg\chi_M^{\mathfrak{p}}=\deg\chi_M^{\mathfrak{q}}$, as desired. ■

Definition 3.9. We write $d(M)$ to denote $\deg\chi_M^{\mathfrak{q}}$ for some $\mathfrak{m}$-primary ideal $\mathfrak{q}$.

THEOREM 3.10. Let $(A,\mathfrak{m})$ be a Noetherian local ring and $0 \to M' \to M \to M'' \to 0$ a short exact sequence of finite A-modules. Then

$$d(M) = \max\{d(M'), d(M'')\}.$$

Further, if $\mathfrak{q}$ is an $\mathfrak{m}$-primary ideal of A, then $\chi^{\mathfrak{q}}_M - \chi^{\mathfrak{q}}_{M''}$ and $\chi^{\mathfrak{q}}_{M'}$ have the same leading coefficient.

Proof. Let $\mathfrak{q}$ be an $\mathfrak{m}$-primary ideal and let $\lambda := \lambda_{A/\mathfrak{q}}$ for brevity. Further, we may assume that $M' \subseteq M$ and $M'' = M/M'$. We have

$$\chi^{\mathfrak{q}}_{M''}(n) = \lambda\left(\frac{M''}{\mathfrak{q}^{n+1}M''}\right) = \lambda\left(\frac{M}{M' + \mathfrak{q}^{n+1}M}\right) = \lambda\left(\frac{M}{\mathfrak{q}^{n+1}M}\right) - \underbrace{\lambda\left(\frac{M' + \mathfrak{q}^{n+1}M}{\mathfrak{q}^{n+1}M}\right)}_{\varphi(n)}.$$

Note that $\varphi(n)$ is a polynomial for $n \gg 0$. Note that

$$\frac{M' + \mathfrak{q}^{n+1}M}{\mathfrak{q}^{n+1}M} = \frac{M'}{M' \cap \mathfrak{q}^{n+1}M},$$

and from the Artin-Rees lemma, we know that the filtration $\left(M' \cap \mathfrak{q}^{n+1}\right)_{n \geqslant 0}$ is $\mathfrak{q}$-stable. In particular, there is an integer $N > 0$ such that whenever $n > N$, we have

$$\mathfrak{q}^{n+1}M' \subseteq M' \cap \mathfrak{q}^{n+1}M = \mathfrak{q}^{n-N}\left(M' \cap \mathfrak{q}^{N+1}M\right) \subseteq \mathfrak{q}^{n-N}M'.$$

This gives $\chi^{\mathfrak{q}}_{M'}(n) \geqslant \varphi(n) \geqslant \chi^{\mathfrak{q}}_{M'}(n-N)$ for all $n > N$. This shows that $\deg \varphi = d(M')$ and φ and $\chi^{\mathfrak{q}}_{M'}$ have the same leading coefficient. The conclusion now follows, since all polynomials involved have positive leading coefficients. ■

COROLLARY 3.11. If $M \twoheadrightarrow N$ is an epimorphism of A-modules, then $d(N) \leqslant d(M)$.

Proof. Apply the theorem to the short exact sequence $0 \to \ker \to M \to N \to 0$. ■

DEFINITION 3.12. Let $(A,\mathfrak{m})$ be a Noetherian local ring and M a finite A-module. We define the *Chevalley dimension* of M to be

$$\delta(M) := \inf\left\{n : \exists\, x_1, \ldots, x_n \in \mathfrak{m} \text{ such that } \lambda_A\left(\frac{M}{(x_1,\ldots,x_n)M}\right) < \infty\right\}.$$

THEOREM 3.13 (DIMENSION THEOREM). Let $(A,\mathfrak{m})$ be a Noetherian local ring and M a finite A-module. Then

$$\dim \operatorname{Supp}_R(M) =: \dim M = d(M) = \delta(M).$$

Proof. We prove this theorem in three steps by proving the sequence of inequalities $\dim M \leqslant d(M) \leqslant \delta(M) \leqslant \dim M$.

Step 1. We show that $d(M) \geqslant \dim M$. To this end, we first establish this inequality for $M = A$ by induction on $d(A)$. If $d(A) = 0$, then $\lambda(A/\mathfrak{m}^n)$ is eventually constant, so that $\mathfrak{m}^n = \mathfrak{m}^{n+1}$ for $n \gg 0$, equivalently, due to Nakayama's lemma, $\mathfrak{m}^n = 0$, that is, $\dim A = 0$. Next, suppose $d(A) > 0$ and consider a strictly ascending chain of prime ideals $\mathfrak{p}_0 \subseteq \mathfrak{p}_1 \subseteq \cdots \subseteq \mathfrak{p}_e$. If $e = 0$, then there is nothing to prove. Suppose now that $e > 0$ and choose $x \in \mathfrak{p}_1 \setminus \mathfrak{p}_0$. Let $B = A/(\mathfrak{p}_0 + xA)$, which is a ring and an A-module fitting into a short exact sequence of A-modules

$$0 \to A/\mathfrak{p}_0 \xrightarrow{\cdot x} A/\mathfrak{p}_0 \to B \to 0.$$

Due to Theorem 3.10, $d_A(B) < d_A(A/\mathfrak{p}_0) \leqslant d_A(A)$. Further, using the induction hypothesis, $\dim B \leqslant d_B(B) = d_A(B) < d_A(A)$. Since the image of the strictly ascending chain of prime ideals $\mathfrak{p}_1 \subseteq \cdots \subseteq \mathfrak{p}_e$ in B is also a strictly ascending chain of prime ideals, $e-1 < d_A(A)$, so that $e \leqslant d_A(A)$. Taking a supremum over all such e's, $\dim A \leqslant d(A)$.

Now, let M be a finite A-module. One can find a filtration

$$0 = M_0 \subseteq M_1 \subseteq \cdots \subseteq M_r = M$$

such that $M_i/M_{i-1} \cong A/\mathfrak{p}_i$ as an A-module for some prime ideal $\mathfrak{p}_i$. Repeatedly invoking Theorem 3.10,

$$d(M) = \sup_{1\leqslant i\leqslant r} d_A(A/\mathfrak{p}_i) = \sup_{1\leqslant i\leqslant r} d_{A/\mathfrak{p}_i}(A/\mathfrak{p}_i) \geqslant \sup_{1\leqslant i\leqslant r} \dim A/\mathfrak{p}_i = \dim M,$$

since $\operatorname{Supp}_R(M) = V(\mathfrak{p}_1) \cup \cdots \cup V(\mathfrak{p}_r)$, as desired.

Step 2. We show that $\delta(M) \geqslant d(M)$. Let $s = \delta(M)$. If $s = 0$, then M is Artinian, so that $\lambda_R(M) < \infty$, i.e., $\chi_M^{\mathfrak{m}}$ is bounded, and hence $d(M) = 0$.

Let $x_1, \ldots, x_s \in \mathfrak{m}$ be such that $\lambda_A\left(\frac{M}{(x_1,\ldots,x_s)M}\right) < \infty$, and set $M_i = M/(x_1,\ldots,x_i)M$. Clearly $\delta(M_i) = s-i$. Further, we have

$$\begin{aligned}\lambda_R(M_1/\mathfrak{m}^n M_1) &= \lambda_R\left(\frac{M}{x_1M + \mathfrak{m}^n M}\right)\\ &= \lambda_R(M/\mathfrak{m}^n M) - \lambda_R\left(\frac{x_1M + \mathfrak{m}^n M}{\mathfrak{m}^n M}\right).\end{aligned}$$

The map $M \xrightarrow{\cdot x_1} M/\mathfrak{m}^n M$ has kernel $(\mathfrak{m}^n M : x_1)$ and image equal to $\frac{x_1M+\mathfrak{m}^n M}{\mathfrak{m}^n M}$, and hence

$$\lambda_R(M_1/\mathfrak{m}^n M_1) = \lambda_R(M/\mathfrak{m}^n M) - \lambda_R\left(\frac{M}{(\mathfrak{m}^n : x_1)M}\right) \geqslant \lambda_R(M/\mathfrak{m}^n M) - \lambda_R(M/\mathfrak{m}^{n-1}M) = \lambda_R\left(\mathfrak{m}^{n-1}M/\mathfrak{m}^n M\right).$$

The right hand side is a polynomial of degree $d(M)-1$ for $n \gg 0$. Thus $d(M_1) \geqslant d(M)-1$. Inductively, $d(M_i) \geqslant d(M) - i$, in particular, $0 = d(M_s) \geqslant d(M) - s$, whence $\delta(M) \geqslant d(M)$.

Step 3. Finally, we show that $\dim M \geqslant \delta(M)$ by induction on $\dim M$. If $\dim M = 0$, then $\operatorname{Supp}_R(M) = \{\mathfrak{m}\}$, so that $\lambda_R(M) < \infty$, i.e., $\delta(M) = 0$. Now suppose $\dim M > 0$ and let $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$ denote the associated primes of M. Since $\dim M > 0$, $\mathfrak{m}$ is not an associated prime, so that by prime avoidance, there is an element $x_1 \in \mathfrak{m} \setminus \bigcup_{i=1}^{r} \mathfrak{p}_r$. Let $M_1 = M/x_1M$, so that $\dim M_1 \leqslant \dim M - 1$, and clearly $\delta(M) \leqslant \delta(M_1) + 1$, so that $\delta(M) \leqslant \dim M$, as desired. ■

Corollary 3.14. Let $(R, \mathfrak{m})$ be a Noetherian local ring, M a finite R-module, and $a \in \mathfrak{m}$ a non-zero-divisor on M. Then $\dim M/aM = \dim M - 1$.

Proof. Let $M' = M/aM$. Then $\delta(M) \leqslant \delta(M') + 1$, so that $\dim M' \leqslant \dim M + 1$. Further, since a is not contained in the union of all associated primes of M, we have

$$\dim \operatorname{Supp}_A(M') = \dim \operatorname{Supp}_A(M) \cap V(a) \leqslant \dim M - 1,$$

so that $\dim M' = \dim M - 1$. ■

Theorem 3.15 (Krull's Hauptidealsatz). Let R be a Noetherian ring and $I = (a_1, \ldots, a_r)$ a proper ideal in R. If $\mathfrak{p}$ is a minimal prime ideal containing I, then $\operatorname{ht}\mathfrak{p} \leqslant r$.

Proof. Note that $IR_{\mathfrak{p}}$ is an $\mathfrak{p}R_{\mathfrak{p}}$-primary ideal, so that $\delta(R_{\mathfrak{p}}) \leqslant r$. Hence, $\operatorname{ht}\mathfrak{p} = \dim R_{\mathfrak{p}} \leqslant r$. ■

THEOREM 3.16. Let $\mathfrak{p}$ be a prime ideal of height $r \geqslant 0$ in a Noetherian ring R.

(1) $\mathfrak{p}$ is a minimal prime over some ideal $(a_1,\dots,a_r)$ generated by r elements.

(2) if $b_1,\dots,b_s \in \mathfrak{p}$, then $\operatorname{ht}\mathfrak{p}/(b_1,\dots,b_s) \geqslant r-s$.

(3) if $a_1,\dots,a_r$ are as in (1), we have

$$\operatorname{ht}\mathfrak{p}/(a_1,\dots,a_i) = r-i \quad \text{for} \quad 1 \leqslant i \leqslant r.$$

Proof. (1) We have $\dim R_{\mathfrak{p}} = \operatorname{ht}\mathfrak{p} = r$, so that there exist $\frac{a_1}{s_1},\dots,\frac{a_r}{s_r} \in R_{\mathfrak{p}}$ generating an $\mathfrak{p}R_{\mathfrak{p}}$-primary ideal. Let $I = (a_1,\dots,a_r)$. Then $IR_{\mathfrak{p}}$ is $\mathfrak{p}R_{\mathfrak{p}}$-primary, so that $\mathfrak{p}$ is a minimal prime over I, as desired.

(2) Set $\overline{R} = A/(b_1,\dots,b_s)$ and $\overline{\mathfrak{p}} = \mathfrak{p}/(b_1,\dots,b_s)$. Let $t = \operatorname{ht}\overline{\mathfrak{p}}$. Due to (1), there exist $\overline{c}_1,\dots,\overline{c}_t \in \overline{R}$ such that $\overline{\mathfrak{p}}$ is a minimal prime containing $(\overline{c}_1,\dots,\overline{c}_t)\overline{R}$. Pick lifts c_i of $\overline{c}_i$. Then $\mathfrak{p}$ is a minimal prime containing $(b_1,\dots,b_s,c_1,\dots,c_t)$. By Theorem 3.15, $s+t \geqslant \operatorname{ht}\mathfrak{p} = r$, as desired.

(3) Note that $\overline{\mathfrak{p}} = \mathfrak{p}/(a_1,\dots,a_i)$ is a minimal prime containing $(\overline{a}_{i+1},\dots,\overline{a}_r)$. Thus, due to Theorem 3.15, $\operatorname{ht}\overline{\mathfrak{p}} \leqslant r-i$. But due to (2), $\operatorname{ht}\overline{\mathfrak{p}} \geqslant r-i$ whence the equality follows. ■

LEMMA 3.17. Let $(A,\mathfrak{m})$ be a Noetherian local ring and $I \subseteq \mathfrak{m}$ an ideal of height $r \geqslant 0$. Then there exists a sequence $\underline{a} = a_1,\dots,a_r \in I$ such that $\operatorname{ht}(a_1,\dots,a_i) = i$ for $1 \leqslant i \leqslant r$.

Proof. For each $0 \leqslant i \leqslant n = \dim A$, let $\mathscr{P}_i$ denote the set of height i primes in A. Using Prime Avoidance, note that I cannot be contained in a finite union of primes in $\bigcup_{i=0}^{r-1} \mathscr{P}_i$. Using this fact, choose $x_1 \in I$ but not contained in any of the primes in $\mathscr{P}_0$ (note that this set is finite). Clearly any prime containing (x_1) must have height at least 1.

Inductively suppose the sequence $x_1,\dots,x_i$ has been constructed. Choose $x_{i+1} \in I$ but not in

$$\operatorname{Ass}_A(A/(x_1,\dots,x_i)) \cap (\mathscr{P}_0 \cup \dots \cup \mathscr{P}_i).$$

If $\mathfrak{p}$ is a minimal prime over $(x_1,\dots,x_{i+1})$, due to Prime Avoidance, it must contain some minimal prime $\mathfrak{q}$ over $(x_1,\dots,x_i)$. If $\operatorname{ht}\mathfrak{q} = i$, then $x_{i+1} \in \mathfrak{p} \setminus \mathfrak{q}$ by construction, and therefore, $\operatorname{ht}\mathfrak{p} \geqslant i+1$. Thus due to the Hauptidealsatz (Theorem 3.15), $\operatorname{ht}(x_1,\dots,x_{i+1}) = i+1$, thereby completing the proof. ■

DEFINITION 3.18. A ring A is said to be *catenary* if for any prime ideals $\mathfrak{p}$ and $\mathfrak{q}$ of A with $\mathfrak{p} \subseteq \mathfrak{q}$, there exists a saturated chain of prime ideals starting from $\mathfrak{p}$ and ending at $\mathfrak{q}$, and all such chains have the same (finite) length.

A ring A is said to be *universally catenary* if every finitely generated A-algebra is catenary.

LEMMA 3.19. A Noetherian ring A is catenary if and only if for every pair of prime ideals $\mathfrak{p} \subseteq \mathfrak{q}$,

$$\operatorname{ht}\mathfrak{q} = \operatorname{ht}\mathfrak{p} + \operatorname{ht}(\mathfrak{q}/\mathfrak{p}).$$

Proof. We prove the converse direction first. Let $\mathfrak{p} = \mathfrak{p}_0 \subsetneq \dots \subsetneq \mathfrak{p}_n = \mathfrak{q}$ be a saturated chain of prime ideals from $\mathfrak{p}$ to $\mathfrak{q}$. Clearly $\operatorname{ht}(\mathfrak{p}_{i+1}/\mathfrak{p}_i) = 1$, so that

$$\operatorname{ht}\mathfrak{q} = \operatorname{ht}\mathfrak{p} + n \implies n = \operatorname{ht}\mathfrak{q} - \operatorname{ht}\mathfrak{p},$$

that is, the length of every saturated chain of primes from $\mathfrak{p}$ to $\mathfrak{q}$ is the same.

Conversely, suppose A is catenary. Choosing a chain of prime ideals ending at $\mathfrak{p}$ of length $\operatorname{ht}\mathfrak{p}$ and appending to it a saturated chain of prime ideals from $\mathfrak{p}$ to $\mathfrak{q}$, we have the inequality:

$$\operatorname{ht}\mathfrak{q} \geqslant \operatorname{ht}\mathfrak{p} + \operatorname{ht}(\mathfrak{q}/\mathfrak{p}).$$

On the other hand, suppose $\mathfrak{p}_0$ is a minimal prime such that there exists a saturated chain of prime ideals from $\mathfrak{p}_0$ to $\mathfrak{q}$ of length $\operatorname{ht}\mathfrak{q}$. Choose a saturated chain of prime ideals from $\mathfrak{p}_0$ to $\mathfrak{p}$ (which has length at most $\operatorname{ht}\mathfrak{p}$) and append to it a saturated chain of prime ideals from $\mathfrak{p}$ to $\mathfrak{q}$. Since A is catenary, this chain must have the same length as the aforementioned chain of length $\operatorname{ht}\mathfrak{q}$ from $\mathfrak{p}_0$ to $\mathfrak{q}$. This shows that

$$\operatorname{ht}\mathfrak{q} \leqslant \operatorname{ht}\mathfrak{p} + \operatorname{ht}(\mathfrak{q}/\mathfrak{p}),$$

whence equality holds. ■

§§ System of Parameters

II. Homological Methods

§4 Regular Sequences

Definition 4.1. Let A be a ring and M an A-module. An element $a \in A$ is said to be *M-regular* if a is a non-zero-divisor on M. A sequence $\underline{a} = a_1, \ldots, a_n$ is said to be an *M-sequence* if the following two conditions hold:

(1) a_i is $M/(a_1, \ldots, a_{i-1})M$-regular for $1 \leqslant i \leqslant n$.

(2) $M \neq (a_1, \ldots, a_n)M$.

Remark 4.2. Note that the permutation of an M-sequence need not be an M-sequence.

Lemma 4.3. If $a_1, \ldots, a_n$ is an M-sequence, and if $a_1\xi_1 + \cdots + a_n\xi_n = 0$ with $\xi_i \in M$ for $1 \leqslant i \leqslant n$, then $\xi_i \in (a_1, \ldots, a_n)M$ for all $1 \leqslant i \leqslant n$.

Proof. Note that $a_n\xi_n = -a_1\xi_1 - \cdots - a_{n-1}\xi_{n-1} \in (a_1, \ldots, a_{n-1})M$, so that $\xi_n \in (a_1, \ldots, a_{n-1})M$, and we can write

$$\xi_n = a_1\eta_1 + \cdots + a_{n-1}\eta_{n-1}$$

for some $\eta_1, \ldots, \eta_{n-1} \in M$. Substituting this back, we have

$$a_1(\xi_1 + a_n\eta_1) + \cdots + a_{n-1}(\xi_{n-1} + a_n\eta_{n-1}) = 0.$$

Repeating our above argument, we have

$$\xi_{n-1} + a_n\eta_{n-1} \in (a_1, \ldots, a_{n-2})M,$$

so that $\xi_{n-1} \in (a_1, \ldots, a_n)M$. Continuing this way, we have our desired conclusion. ■

Theorem 4.4. If $a_1, \ldots, a_n$ is an M-sequence, then so is $a_1^{m_1}, \ldots, a_n^{m_n}$ for any positive integers $m_1, \ldots, m_n$.

Proof. Note that it suffices to show that for any positive integer m, $a_1^m, a_2, \ldots, a_n$ is an M-sequence. We shall prove this statement by induction on $m \geqslant 1$. The base case with $m = 1$ is trivial. Since a_1 is M-regular, so is a_1^m. Suppose for $i > 1$, we have the relation

$$a_i\omega = a_1^m\xi_1 + \cdots + a_{i-1}\xi_{i-1} \quad \text{with} \quad \xi_j \in M.$$

Due to the induction hypothesis, we know that $a_1^{m-1}, a_2, \ldots, a_i$ is an M-sequence, so that by Lemma 4.3, there exist $\eta_1, \ldots, \eta_{i-1} \in M$ such that

$$\omega = a_1^{m-1}\eta_1 + a_2\eta_2 + \cdots + a_{i-1}\eta_{i-1}.$$

Hence, we obtain

$$0 = a_1^{m-1}(a_1\xi_1 - a_i\eta_1) + a_2(\xi_2 - a_i\eta_2) + \cdots + a_{i-1}(\xi_{i-1} - a_i\eta_{i-1}).$$

Invoking Lemma 4.3 again,

$$a_1\xi_1 - a_i\eta_1 \in (a_1^{m-1}, a_2, \ldots, a_{i-1})M,$$

and hence,

$$a_i\eta_1 \in (a_1,\dots,a_{i-1})M.$$

But since $a_1,\dots,a_i$ is an M-sequence,

$$\eta_1 \in (a_1,\dots,a_{i-1})M.$$

This shows that $\omega \in (a_1^m, a_2,\dots,a_{i-1})M$, as desired. ■

Definition 4.5. Let A be a ring and $x_1,\dots,x_n \in A$. We define a complex $K_\bullet$ as follows: set $K_0 = A$, and $K_p = 0$ if $p < 0$ or $p > n$. For $1 \leqslant p \leqslant n$, set

$$K_p = \bigoplus_{1\leqslant i_1<\cdots<i_p\leqslant n} Ae_{i_1,\dots,i_p}.$$

The differential $\mathrm{d}\colon K_p \to K_{p-1}$ is defined by

$$\mathrm{d}\left(e_{i_1,\dots,i_p}\right) = \sum_{k=1}^{p}(-1)^{k-1}x_{i_k}e_{i_1,\dots,\widehat{i_k},\dots,i_p} \in K_{p-1}$$

for $1 \leqslant p \leqslant n$, and $\mathrm{d} = 0$ for $p \leqslant 0$ or $p > n$. This is called the *Koszul complex* corresponding to the sequence $\underline{x} = x_1,\dots,x_n$ and is often denoted by $K(\underline{x})_\bullet$. If M is an A-module, we set $K(\underline{x},M)_\bullet = K(\underline{x})_\bullet \otimes_A M$.

Interlude 4.6 (Tensor Product of Complexes). Let $(A_\bullet,\mathrm{d})$ and $\left(B_\bullet,\mathrm{d}'\right)$ be two complexes of R-modules. We define the complex $(A\otimes_R B)_\bullet$ as

$$(A\otimes_R B)_p = \bigoplus_{i+j=p} A_i \otimes_R B_j,$$

with the differential given by

$$\mathrm{d}\left(a_i\otimes b_j\right) = \mathrm{d}a_i \otimes b_j + (-1)^i a_i \otimes \mathrm{d}b_j$$

where $a_i \in A_i$ and $b_j \in B_j$ for some integers i and j.

Remark 4.7. It is straightforward to check that $K(\underline{x})_\bullet$ is a complex, i.e., $\mathrm{d}^2 = 0$. Further, for each $x \in A$, consider the complex $C(x)_\bullet$ given by

$$\cdots \to 0 \to A \xrightarrow{\cdot x} A \to 0 \to \cdots$$

with A in degrees 0 and 1. One can then check that

$$K(\underline{x}) \cong C(x_1)\otimes_A \cdots \otimes_A C(x_n).$$

Since the tensor product of complexes is commutative (up to isomorphism), we note that permuting the sequence $\underline{x} = x_1,\dots,x_n$ results in an isomorphic complex.

Definition 4.8. Let M be an A-module and $\underline{x} = x_1,\dots,x_n$ be a sequence in M. We define the *Koszul homology modules* to be the homology modules of the complex $K(\underline{x},M)_\bullet$ and are denoted $H_i\left(\underline{x},M\right)$ for all $i \in \mathbb{Z}$.

It is straightforward to see that

$$H_0(\underline{x},M) \cong M/(x_1,\dots,x_n)M \quad \text{and} \quad H_n(\underline{x},M) = \left(0 :_M \underline{x}\right) = \{\xi \in M : x_1\xi = \cdots = x_n\xi = 0\}.$$

INTERLUDE 4.9 (LONG EXACT SEQUENCE). Let $0 \to M' \to M \to M'' \to 0$ be a short exact sequence of A-modules, and $\underline{x} = x_1, \ldots, x_n \in A$ a sequence. Then there is an associated long exact sequence in homology:

$$\cdots \to H_i(\underline{x}, M') \to H_i(\underline{x}, M) \to H_i(\underline{x}, M'') \to H_{i+1}(\underline{x}, M') \to \cdots.$$

Indeed, consider the following commutative diagram of A-modules:

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & \\
\cdots \arrow[r] & K_p \otimes_A M' \arrow[r] \arrow[d] & K_{p-1} \otimes_A M' \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & K_p \otimes_A M \arrow[r] \arrow[d] & K_{p-1} \otimes_A M \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & K_p \otimes_A M'' \arrow[r] \arrow[d] & K_{p-1} \otimes_A M'' \arrow[r] \arrow[d] & \cdots \\
 & 0 & 0 &
\end{tikzcd}.$$

Since each K_p is a free A-module, it is A-flat, so that all vertical columns are exact. This short exact sequence of complexes induces the desired long exact sequence of homology modules.

For a complex $C_\bullet$ and $x \in A$, let $C(x)_\bullet$ denote the complex $C_\bullet \otimes K(x)_\bullet$.

THEOREM 4.10. Let $C_\bullet$ be a complex of A-modules and $x \in A$. Then there is an exact sequence of complexes

$$0 \to C_\bullet \to C(x)_\bullet \to \widetilde{C}_\bullet \to 0,$$

where $\widetilde{C}_\bullet$ is given by $\widetilde{C}_p = C_{p+1}$ with the same differential. This short exact sequence gives rise to a corresponding homology long exact sequence:

$$\cdots \to H_p(C_\bullet) \to H_p(C(x)_\bullet) \to H_{p-1}(C_\bullet) = H_p\left(\widetilde{C}_\bullet\right) \xrightarrow{(-1)^{p-1}x} H_{p-1}(C_\bullet) \to \cdots.$$

Further, we have $x \cdot H_p(C(x)_\bullet) = 0$ for all $p \in \mathbb{Z}$.

Proof. Let d denote the differential of the complex $C_\bullet$. The complex $K(x)_\bullet$ is given by

$$\cdots \to 0 \to Ae_1 \xrightarrow{e_1 \mapsto x} A \to 0 \to \cdots,$$

so that we may identify

$$C(x)_p = C_p \oplus C_{p-1},$$

with the differential

$$\widetilde{\mathrm{d}}\left(\xi, \eta\right) = \left(\mathrm{d}\xi + (-1)^{p-1} x\eta, \mathrm{d}\eta\right).$$

Consider the short exact sequence of complexes:

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & \\
\cdots \arrow[r] & C_p \arrow[r] \arrow[d] & C_{p-1} \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & C_p \oplus C_{p-1} \arrow[r] \arrow[d] & C_{p-1} \oplus C_{p-2} \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & C_{p-1} \arrow[r] \arrow[d] & C_{p-2} \arrow[r] \arrow[d] & \cdots \\
 & 0 & 0 &
\end{tikzcd}$$

where the vertical maps are the canonical inclusions and the projections. That these maps are indeed chain maps is easy to check.

It remains to compute the boundary map $H_p(\widetilde{C}_\bullet) \to H_{p-1}(C_\bullet)$. Let $\eta \in \widetilde{C}_p = C_{p-1}$ be such that $\mathrm{d}\eta = 0$. Consider $(0,\eta) \in C_p \oplus C_{p-1}$ whose image under $\widetilde{\mathrm{d}}$ is $\left((-1)^{p-1}x\eta, 0\right)$, which lifts to $(-1)^{p-1}x\eta \in C_{p-1}$. Thus, the boundary map $H_{p-1}(C_\bullet) \to H_{p-1}(C_\bullet)$ is multiplication by $(-1)^{p-1}x$.

Finally, we show that x annihilates $H_p(C(x)_\bullet)$. Indeed, let $(\xi,\eta) \in C_p \oplus C_{p-1}$ be such that $\widetilde{\mathrm{d}}(\xi,\eta) = 0$, equivalently, $\mathrm{d}\xi = (-1)^p x\eta$ and $\mathrm{d}\eta = 0$. Thus

$$x \cdot (\xi,\eta) = (x\xi, x\eta) = \mathrm{d}\left(0, (-1)^p \xi\right),$$

consequently, x annihilates $H_p(C(x)_\bullet)$ for all $p \in \mathbb{Z}$. ■

Corollary 4.11. Let $\underline{x} = x_1,\dots,x_n$ be a sequence in A. Then $(x_1,\dots,x_n)$ annihilates $H_p(\underline{x},M)$ for all $p \in \mathbb{Z}$.

Proof. This follows from the fact that the Koszul complex is precisely $C(x_1) \otimes_A \cdots \otimes_A C(x_n) \otimes_A M$. ■

Theorem 4.12. Let A be a ring, M an A-module, and $\underline{x} = x_1,\dots,x_n \in A$.

(1) If $\underline{x}$ is an M-sequence, then $H_p(\underline{x},M) = 0$ for all $p \neq 0$ and $H_0(\underline{x},M) = M/(\underline{x})M$.

(2) Suppose that $(A,\mathfrak{m})$ is a local ring, $x_1,\dots,x_n \in \mathfrak{m}$, and M is a finitely presented A-module. If $H_1(\underline{x},M) = 0$ and $M \neq 0$, then $\underline{x}$ is an M-sequence.

Proof. We prove both statements by induction on $n \geqslant 1$.

(1) The base case with $n = 1$ is trivial, since $H_1(x_1,M) = (0 :_M x_1) = 0$ since x_1 is M-regular. Suppose now that $n > 1$. If $p > 1$, using Theorem 4.10 we have a long exact sequence:

$$0 = H_p(x_1,\dots,x_{n-1},M) \to H_p(x_1,\dots,x_n,M) \to H_{p-1}(x_1,\dots,x_{n-1},M) = 0$$

, where the two terms are zero from the inductive hypothesis. It follows that $H_p(x_1,\dots,x_n,M) = 0$ too. Next, if $p = 1$, then due to Theorem 4.10, we have an exact sequence

$$0 = H_1(x_1,\dots,x_{n-1},M) \to H_1(x_1,\dots,x_n,M) \to H_0(x_1,\dots,x_{n-1},M) = M/(x_1,\dots,x_{n-1})M \xrightarrow{\cdot x_n} M/(x_1,\dots,x_{n-1})M.$$

Since x_n is $M/(x_1,\dots,x_{n-1})M$-regular, the last map in the above sequence is injective, so that $H_1(x_1,\dots,x_n,M) = 0$, as desired.

(2) Again, the base case with $n = 1$ is trival since $H_1(x_1, M) = (0 :_M x_1)$, which is zero if and only if x_1 is M-regular.

Since M is a finitely presented, all Koszul homology groups $K(\underline{x}, M)$ of M must be finitely generated. The long exact sequence furnished by Theorem 4.10 is of the form

$$H_1(x_1, \ldots, x_{n-1}, M) \xrightarrow{\cdot(-x_n)} H_1(x_1, \ldots, x_{n-1}, M) \to H_1(x_1, \ldots, x_n, M) = 0.$$

Thus the multiplication map by x_n on $H_1(x_1, \ldots, x_{n-1}, M)$ is surjective. Since $x_n \in \mathfrak{m}$ and the Koszul homology is finitely generated, we must have that $H_1(x_1, \ldots, x_{n-1}, M) = 0$ by Nakayama's lemma. Due to the inductive hypothesis, we must have that $x_1, \ldots, x_{n-1}$ is an M-sequence. Finally, due to Theorem 4.10, there is an exact sequence

$$0 = H_1(x_1, \ldots, x_n, M) \to H_0(x_1, \ldots, x_{n-1}, M) = M/(x_1, \ldots, x_{n-1})M \xrightarrow{\cdot x_n} M/(x_1, \ldots, x_{n-1})M,$$

so that the map $M/(x_1, \ldots, x_{n-1})M \xrightarrow{\cdot x_n} M/(x_1, \ldots, x_{n-1})M$ is injective, that is, x_n is $M/(x_1, \ldots, x_{n-1})M$-regular. ■

THEOREM 4.13. Let A be a Noetherian ring, M a finite A-module, and I an ideal in A with $IM \neq M$. For a positive integer $n > 0$, the following are equivalent:

(1) $\mathrm{Ext}^i_A(N, M) = 0$ for all $i < n$ and for any finite A-module N with $\mathrm{Supp}_A(N) \subseteq V(I)$.

(2) $\mathrm{Ext}^i_A(A/I, M) = 0$ for all $i < n$.

(3) $\mathrm{Ext}^i_A(N, M) = 0$ for all $i < n$ and for some finite A-module N with $\mathrm{Supp}_A(N) = V(I)$.

(4) there exists an M-sequence of length n contained in I.

Proof. The implications (1) $\Longrightarrow$ (2) $\Longrightarrow$ (3) are clear.
(3) $\Longrightarrow$ (4) We shall prove this statement by induction on n. For the base case, suppose that $n = 1$, that is, $\mathrm{Hom}_A(N, M) = 0$ where N is a finite A-module with $\mathrm{Supp}_A(N) = V(I)$. If I does not contain a non-zero-divisor on M, then

$$I \subseteq \bigcup_{\mathfrak{p} \in \mathrm{Ass}_A(M)} \mathfrak{p},$$

so that by Prime Avoidance, there exists an associated prime $\mathfrak{p} \in \mathrm{Ass}_A(M)$ such that $I \subseteq \mathfrak{p}$ – in particular, $\mathfrak{p} \in V(I) = \mathrm{Supp}_A(N)$, that is, $N_\mathfrak{p} \neq 0$. Due to Nakayama's lemma, this is equivalent to $N_\mathfrak{p} \otimes_{A_\mathfrak{p}} \kappa(\mathfrak{p}) \neq 0$. Thus, there is a non-zero $\kappa(\mathfrak{p})$-linear, and hence $A_\mathfrak{p}$-linear map $N_\mathfrak{p} \otimes_{A_\mathfrak{p}} \kappa(\mathfrak{p}) \to \kappa(\mathfrak{p})$. Further, since $\mathfrak{p} \in \mathrm{Ass}_A(M)$, there is an injective A-linear map $A/\mathfrak{p} \hookrightarrow M$, which localizes into an injective $A_\mathfrak{p}$-linear map $\kappa(\mathfrak{p}) \hookrightarrow M_\mathfrak{p}$. As a result, the composite:

$$N_\mathfrak{p} \twoheadrightarrow N_\mathfrak{p} \otimes_{A_\mathfrak{p}} \kappa(\mathfrak{p}) \to \kappa(\mathfrak{p}) \hookrightarrow M_\mathfrak{p}$$

is non-zero. That is,

$$0 \neq \mathrm{Hom}_{A_\mathfrak{p}}\left(N_\mathfrak{p}, M_\mathfrak{p}\right) = (\mathrm{Hom}_A(N, M))_\mathfrak{p},$$

whence $\mathrm{Ext}^0_A(N, M) = \mathrm{Hom}_A(N, M) \neq 0$, a contradiction. Thus I must contain an M-regular element. This commpletes our proof of the base case.

Suppose now that $n > 1$. We know again that $\mathrm{Ext}^i_A(N, M) = 0$ for $i < 1$, so by the above argument, I contains an M-regular element, say x_1. Set $M_1 = M/x_1M$. There is a short exact sequence

$$0 \to M \xrightarrow{x_1} M \to M_1 \to 0.$$

This gives rise to a long exact sequence:

$$\cdots \operatorname{Ext}_A^i(N,M) \xrightarrow{x_1} \operatorname{Ext}_A^i(N,M) \to \operatorname{Ext}_A^i(N,M_1) \to \operatorname{Ext}^{i+1}(N,M_1) \to \cdots.$$

If $i < n-1$, then $\operatorname{Ext}_A^i(N,M) = \operatorname{Ext}_A^{i+1}(N,M) = 0$, so that $\operatorname{Ext}_A^i(N,M) = 0$. Due to the inductive hypothesis, there exists an M_1-sequence $x_2,\ldots,x_n$ contained in I. Hence, $x_1,\ldots,x_n$ is an M-sequence contained in I.

(4) $\Longrightarrow$ (1) We shall proceed by induction on $n \geqslant 1$. Consider the base case $n = 1$, that is, I contains an M-regular element x, and $\operatorname{Supp}_A(N) \subseteq V(I)$. Since N is a finite A-module, $\operatorname{Supp}_A(N) = V(\operatorname{Ann}_A(N))$, so that $I \subseteq \sqrt{\operatorname{Ann}_A(N)}$. Now x^r annihilates N for some $r > 0$ but x^r is still M-regular (Theorem 4.4). This immediately implies that $\operatorname{Ext}_A^0(N,M) = \operatorname{Hom}_A(N,M) = 0$, thereby establishing the base case.

Suppose now that $n > 1$, and let $x_1,\ldots,x_n \in I$ be an M-sequence. Set $M_1 = M/x_1M$ and consider the short exact sequence

$$0 \to M \xrightarrow{x_1} M \to M_1 \to 0,$$

which gives rise to the long exact sequence

$$\cdots \to \operatorname{Ext}_A^{n-2}(N,M_1) \to \operatorname{Ext}_A^{n-1}(N,M) \xrightarrow{x_1} \operatorname{Ext}_A^{n-1}(N,M) \to \cdots.$$

Since I contains an M_1-sequence of length $n-1$, due to the inductive hypothesis, $\operatorname{Ext}_A^{n-2}(N,M) = 0$, that is, multiplication by x_1 induces an injective map on $\operatorname{Ext}_A^{n-1}(N,M)$. Now, as we have observed above, $I \subseteq \sqrt{\operatorname{Ann}_A(N)}$, so that there is a positive integer $r > 0$ such that x_1^r annihilates N, and hence must annihilate $\operatorname{Ext}_A^{n-1}(N,M)$. This is possible if and only if $\operatorname{Ext}_A^{n-1}(N,M) = 0$, as desired. ■

Lemma 4.14. Let I be an ideal in a ring A, M an A-module, and $\underline{a} = a_1,\ldots,a_n \in I$ an M-sequence. Set $M_i = M/(a_1,\ldots,a_i)M$. Then we have the isomorphisms

$$\operatorname{Hom}_A(A/I,M_n) = \operatorname{Ext}_A^0(A/I,M_n) \cong \operatorname{Ext}_A^1(A/I,M_{n-1}) \cong \cdots \cong \operatorname{Ext}_A^n(A/I,M).$$

Proof. More generally, we shall show that if $a \in I$ is M-regular and $M_1 = M/aM$, then $\operatorname{Ext}_A^i(A/I,M) \cong \operatorname{Ext}^{i-1}(A/I,M_1)$ for all i. Clearly our desired statement would follow immediately. Indeed, consider the long exact sequence corresponding to the short exact sequence $0 \to M \xrightarrow{a} M \to M_1 \to 0$:

$$\cdots \to \operatorname{Ext}_A^i(A/I,M) \xrightarrow{a} \operatorname{Ext}_A^i(A/I,M) \to \operatorname{Ext}_A^i(A/I,M_1) \xrightarrow{\delta} \operatorname{Ext}_A^{i+1}(A/I,M) \xrightarrow{a} \operatorname{Ext}_A^{i+1}(A/I,M) \to \cdots.$$

Since $a \in I$, it annihilates A/I, and therefore annihilates $\operatorname{Ext}_A^j(A/I,M)$ for all j. As a result, the connecting morphism δ is both injective and surjective, whence an isomorphism. ■

Theorem 4.15 (Rees). Let A be a Noetherian ring, I an ideal of A, and M a finite A-module such that $IM \neq M$. Then the length of any maximal M-sequence contained in I is a well-determined integer given by

$$\operatorname{Ext}_A^i(A/I,M) = 0 \quad \text{for } i < n \quad \text{and} \quad \operatorname{Ext}_A^n(A/I,M) \neq 0.$$

Proof. First, since A is Noetherian, clearly the length of any maximal M-sequence contained in I is finite. Thus, due to Theorem 4.13, there must exist an index j with $\operatorname{Ext}_A^j(A/I,M) \neq 0$. Let n be the smallest such index. Then due to Theorem 4.13, the length of any M-sequence contained in I is at most n. Conversely, let $a_1,\ldots,a_s$ be a maximal M-sequence contained in I. Clearly $s \leqslant n$. Since this sequence is maximal, every element of I is a non-zero-divisor on $M_s = M/(a_1,\ldots,a_s)M$, whence

$$I \subseteq \bigcup_{\mathfrak{p} \in \operatorname{Ass}_A(M_s)} \mathfrak{p},$$

so that by Prime Avoidance, $I \subseteq \mathfrak{p}$ for some $\mathfrak{p} \in \operatorname{Ass}_A(M_s)$.

Further, due to Lemma 4.14,

$$\operatorname{Ext}_A^s(A/I, M) = \operatorname{Hom}_A\left(A/I, \frac{M}{(a_1,\dots,a_s)M}\right) \neq 0,$$

since we may obtain the non-zero composite map

$$A/I \twoheadrightarrow A/\mathfrak{p} \hookrightarrow M.$$

This shows that $n \leqslant s$, and the conclusion follows. ■

Definition 4.16. Let A be a Noetherian ring, M a finite A-module, and I an ideal of A with $IM \neq M$. We define $\operatorname{depth}(I, M)$ to be the length of a maximal M-sequence contained in I and call it the *I-depth of M*. If $IM = M$, we set $\operatorname{depth}(I, M) = \infty$.

Remark 4.17. (i) Using Theorem 4.13, it is clear that if I and J are two ideal in A with $IM \neq M$, $JM \neq M$, and $V(I) = V(J)$, then $\operatorname{depth}(I, M) = \operatorname{depth}(J, M)$.

(ii) Suppose $\mathfrak{a} \subseteq \operatorname{Ann}_A(M)$ is an ideal, and set $\overline{A} = A/\mathfrak{a}$. Clearly, $a_1,\dots,a_r$ is an M-sequence if and only if the image sequence $\overline{a}_1,\dots,\overline{a}_r$ in $\overline{A}$ is an M-sequence. Therefore, if I is an ideal of I and $\overline{I}$ its image in $\overline{A}$, then $\operatorname{depth}_A(I, M) = \operatorname{depth}_{\overline{A}}(\overline{I}, M)$. In particular, this also shows that $\operatorname{depth}_A(I, M) = \operatorname{depth}_A(I + \mathfrak{a}, M)$.

Theorem 4.18 (Depth Sensitivity of Koszul Complex). Let A be a Noetherian ring, $I = (y_1,\dots,y_n) = (\underline{y})$ an ideal in A, and M a finite A-module such that $IM \neq M$. Set

$$q = \left\{ i : H_i(\underline{y}, M) \neq 0 \right\}.$$

Then $\operatorname{depth}(I, M) = n - q$.

Proof. Let $s = \operatorname{depth}(I, M)$. We shall prove this statement by induction on $s \geqslant 0$. Suppose first that $s = 0$. As we have argued multiple times above, I must be contained in an associated prime $\mathfrak{p} \in \operatorname{Ass}_A(M)$. Hence, there exists $0 \neq \xi \in M$ such that $\mathfrak{p} = \operatorname{Ann}_A(\xi)$. Hence,

$$H_n\left(\underline{y}, M\right) \cong (0 :_M I) \supseteq (0 :_M \mathfrak{p}) \neq 0,$$

i.e., $q = n$ in this case.

Suppose now that $s > 0$, and let $\underline{x} = x_1,\dots,x_s \in I$ be a maximal M-sequnce contained in I. Set $M_1 = M/x_1M$, so that $\operatorname{depth}(I, M_1) = s - 1$ and there is a short exact sequence

$$0 \to M \xrightarrow{x_1} M \to M_1 \to 0.$$

This induces a long exact sequence of Koszul homology as seen in Interlude 4.9:

$$\cdots \to H_i(\underline{x}, M) \xrightarrow{x_1} H_i(\underline{x}, M) \to H_i(\underline{x}, M_1) \to H_{i-1}(\underline{x}, M) \xrightarrow{x_1} H_{i-1}(\underline{x}, M) \to \cdots.$$

Since x_1 annihilates all Koszul homology groups above, we see that the above long exact sequence breaks up into short exact sequences of the form

$$0 \to H_i(\underline{x}, M) \to H_i(\underline{x}, M_1) \to H_{i-1}(\underline{x}, M) \to 0.$$

We note that $H_q(\underline{x},M) \neq 0$, whence due to the above short exact sequence, we must have that $H_{q+1}(\underline{x},M_1) \neq 0$. But similarly, if $i > q+1$, then $H_{i-1}(\underline{x},M) = 0$, so that $H_i(\underline{x},M_1) = H_i(\underline{x},M) = 0$. Thus

$$\sup\left\{i : H_i(\underline{y},M_1) \neq 0\right\} = q+1,$$

so that by the inductive hypothesis,

$$n = (q+1)+(s-1) = q+s,$$

as desired. ■

Remark 4.19. In other words, $\operatorname{depth}(I,M)$ is precisely the number of successive zero terms from the left in the sequence of modules:

$$H_n(\underline{y},M),\ H_{n-1}(\underline{y},M),\ \cdots,\ H_0(\underline{y},M) = M/IM \neq 0.$$

§5 Cohen-Macaulay Rings and Modules

Theorem 5.1 (Ischebeck). Let $(A,\mathfrak{m})$ be a Noetherian local ring, M and N non-zero finite A-modules, and let $k = \operatorname{depth} M$ and $r = \dim N$. Then

$$\operatorname{Ext}^i_A(N,M) = 0 \quad \text{for } i < k-r.$$

Proof. We shall prove this statement by induction on $r \geqslant 0$. If $r = 0$, then $\operatorname{Supp}_A(N) = \{\mathfrak{m}\}$, so that by Theorem 4.13, $\operatorname{Ext}^i_A(N,M) = 0$ for all $i < \operatorname{depth} M$.

Suppose now that $r > 0$. There is a filtration

$$0 = N_0 \subseteq \cdots \subseteq N_n = N,$$

such that $N_i/N_{i-1} = A/\mathfrak{p}_i$ for some prime ideal $\mathfrak{p}_i$ in A. Since N_i/N_{i-1} is a subquotient of N, we have $\dim N_i/N_{i-1} \leqslant r$, so that it suffices to show $\operatorname{Ext}^j_A(A/\mathfrak{p}_i,M) = 0$ for $j < k-r$ and for all i.

Let $N = A/\mathfrak{p}_i$. Since $r > 0$, we may choose an element $x \in \mathfrak{m} \setminus \mathfrak{p}_i$ whence we have an exact sequence

$$0 \to N \xrightarrow{x} N \to N/xN \to 0.$$

Note that $\dim N/xN \leqslant r-1$, so that by the inductive hypothesis, we know that $\operatorname{Ext}^j_A(N/xN,M) = 0$ for all $j < k-(r-1)$. Thus for $j < k-r$, we have the exact sequence:

$$0 = \operatorname{Ext}^j_A(N/xN,M) \to \operatorname{Ext}^j_A(N,M) \xrightarrow{x} \operatorname{Ext}^j_A(N,M) \to \operatorname{Ext}^{j+1}_A(N/xN,M) = 0,$$

so that multiplication by x induces a surjective map on $\operatorname{Ext}^j_A(N,M)$; but since $x \in \mathfrak{m}$, due to Nakayama's lemma, $\operatorname{Ext}^j_A(N,M) = 0$, as desired. ■

Corollary 5.2. Let A be a Noetherian local ring, M a finite A-module, and $\mathfrak{p} \in \operatorname{Ass}_A(M)$. Then $\dim A/\mathfrak{p} \geqslant \operatorname{depth} M$.

Proof. If $\mathfrak{p} \in \operatorname{Ass}_A(M)$, then $\operatorname{Ext}^0_A(A/\mathfrak{p},M) = \operatorname{Hom}_A(A/\mathfrak{p},0) \neq 0$, so that by Theorem 5.1, we must have $\dim A/\mathfrak{p} \geqslant \operatorname{depth} M$. ■

Theorem 5.3. Let $(A,\mathfrak{m},k)$ be a Noetherian local ring, and M a finite A-module. We say that M is a *Cohen-Macaulay (CM) module* if $\operatorname{depth} M = \dim M$, or $M = 0$. If A is a Cohen-Macaulay module over itself, then A is said to be a *Cohen-Macaulay ring*.

THEOREM 5.4. Let A be a Noetherian local ring and M a finite A-module.

(1) If M is a CM A-module, then for any $\mathfrak{p} \in \operatorname{Ass}_A(M)$, $\dim A/\mathfrak{p} = \dim M = \operatorname{depth} M$. In particular, M has no embedded associated primes.

(2) Let $a_1, \ldots, a_r \in \mathfrak{m}$ be an M-sequence and set $M' = M/(a_1, \ldots, a_r)M$. Then

$$M \text{ is a CM } A\text{-module} \iff M' \text{ is.}$$

(3) If M is a CM A-module, then $M_\mathfrak{p}$ is a CM $A_\mathfrak{p}$-module for every prime ideal $\mathfrak{p}$ in A. Further, if $M_\mathfrak{p} \neq 0$, then

$$\operatorname{depth}(\mathfrak{p}, M) = \operatorname{depth}_{A_\mathfrak{p}} M_\mathfrak{p}.$$

Proof. (1) For $\mathfrak{p} \in \operatorname{Ass}_A(M)$, using Corollary 5.2, we have the inequalities

$$\dim M = \dim \operatorname{Supp}_A(M) \geqslant \dim A/\mathfrak{p} \geqslant \operatorname{depth} M,$$

so that $\dim A/\mathfrak{p} = \operatorname{depth} M$.

(2) Clearly $\operatorname{depth} M' = \operatorname{depth} M - r$. Due to Corollary 3.14, we know that $\dim M' = \dim M - r$, whence the conclusion follows.

(3) It suffices to prove the statement in the case $\mathfrak{p} \in \operatorname{Supp}_A(M) = V(\operatorname{Ann}_A(M))$. Note that we have the inequalities:

$$\dim_{A_\mathfrak{p}} M_\mathfrak{p} \geqslant \operatorname{depth}_{A_\mathfrak{p}} M_\mathfrak{p} \geqslant \operatorname{depth}(\mathfrak{p}, M),$$

where the second inequality follows from the fact that the image of any M-sequence in $\mathfrak{p}A_\mathfrak{p}$ is an $M_\mathfrak{p}$-sequence. Hence, it suffices to show that $\dim M_\mathfrak{p} = \operatorname{depth}(\mathfrak{p}, M)$.

We prove this by induction on $\operatorname{depth}(\mathfrak{p}, M)$. If $\operatorname{depth}(\mathfrak{p}, M) = 0$, then $\mathfrak{p}$ is contained in one of the associated primes of M. But since M has no associated primes, all elements of $\operatorname{Ass}_A(M)$ must be the minimal elements of $\operatorname{Supp}_A(M)$[2], and hence $\mathfrak{p}$ must be an associated prime – and in particular, a minimal prime containing $\operatorname{Ann}_A(M)$. Therefore, $\dim M_\mathfrak{p} = 0$, since $\operatorname{Supp}_{A_\mathfrak{p}}(M_\mathfrak{p}) = \{\mathfrak{p}A_\mathfrak{p}\}$.

Next, suppose $\operatorname{depth}(\mathfrak{p}, M) > 0$, choose an M-regular element $a \in \mathfrak{p}$, and set $M' = M/aM$. Clearly

$$\operatorname{depth}(\mathfrak{p}, M') = \operatorname{depth}(\mathfrak{p}, M) - 1,$$

and due to Corollary 3.14,

$$\dim (M')_\mathfrak{p} = \dim M_\mathfrak{p}/aM_\mathfrak{p} = \dim M_\mathfrak{p} - 1.$$

Using the inductive hypothesis, we conclude that $\dim M_\mathfrak{p} = \operatorname{depth}(\mathfrak{p}, M)$, thereby completing the proof. ■

THEOREM 5.5. Let $(A, \mathfrak{m})$ be a CM local ring. For a sequence $\underline{a} = a_1, \ldots, a_r \in \mathfrak{m}$, the following are equivalent:

(1) $\underline{a}$ is an A-sequence.

(2) $\operatorname{ht}(a_1, \ldots, a_i) = i$ for all $1 \leqslant i \leqslant r$.

[2] Recall that the minimal elements of $\operatorname{Ass}_A(M)$ and $\operatorname{Supp}_A(M)$ are the same.

(3) $\operatorname{ht}(a_1,\dots,a_r) = r$.

(4) $a_1,\dots,a_r$ is a part of a system of parameters for A.

Proof. (1) $\Longrightarrow$ (2) We first show that

$$0 < \operatorname{ht}(a_1) < \operatorname{ht}(a_1,a_2) < \cdots < \operatorname{ht}(a_1,\dots,a_r).$$

Indeed, suppose $\operatorname{ht}(a_1,\dots,a_{i-1}) = \operatorname{ht}(a_1,\dots,a_i)$ for some $i \geqslant 1$. Let $\mathfrak{p}_1,\dots,\mathfrak{p}_s$ and $\mathfrak{q}_1,\dots,\mathfrak{q}_t$ denote the minimal primes containing $(a_1,\dots,a_{i-1})$ and $(a_1,\dots,a_i)$ respectively. Then we have the inclusion:

$$\mathfrak{p}_1 \cap \cdots \cap \mathfrak{p}_s = \sqrt{(a_1,\dots,a_{i-1})} \subseteq \sqrt{(a_1,\dots,a_i)} \subseteq \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_t.$$

Since a_i is a non-zero-divisor on $A/(a_1,\dots,a_{i-1})$, it cannot be contained in any of the $\mathfrak{p}_j$'s. Further, by Prime Avoidance, each $\mathfrak{q}_k$ must contain some $\mathfrak{p}_j$, so that

$$\operatorname{ht}\mathfrak{q}_k \geqslant \operatorname{ht}\mathfrak{p}_j + 1 \geqslant \operatorname{ht}(a_1,\dots,a_{i-1}) + 1.$$

This shows that $\operatorname{ht}(a_1,\dots,a_i) \geqslant \operatorname{ht}(a_1,\dots,a_{i-1}) + 1$. Finally, due to the Hauptidealsatz (Theorem 3.15), $\operatorname{ht}(a_1,\dots,a_i) \leqslant i$, whence equality must hold.

(2) $\Longrightarrow$ (3) is clear.

(3) $\Longrightarrow$ (4) If $\dim A = r$, then $\underline{a}$ is a system of parameters. If $\dim A > r$, then $\mathfrak{m}$ is not an associated prime of $(a_1,\dots,a_r)$ due to the Hauptidealsatz (Theorem 3.15). Hence, we may choose $a_{r+1} \in \mathfrak{m}$ that is not contained in any of the minimal primes (equiv. associated primes) of $(a_1,\dots,a_r)$. As we have argued above, $\operatorname{ht}(a_1,\dots,a_{r+1}) = r+1$. Continuing in this fashion, we may extend $\underline{a}$ to a system of parameters for A.

(4) $\Longrightarrow$ (1) It suffices to show that any system of parameters $\underline{x} = x_1,\dots,x_n$ for a CM local ring A is also an A-sequence. We shall do so by inducting on $n = \dim A$. If $\mathfrak{p}$ is a minimal prime in A (equiv. an associated prime of A), then due to Theorem 5.4, $\dim A/\mathfrak{p} = n$. It follows hence that $x_1 \notin \mathfrak{p}$, else $A/\mathfrak{p}$ would be a quotient of A/x_1, which is $(n-1)$-dimensional. Thus, x_1 is A-regular. Set $A' = A/x_1$, which, due to Theorem 5.4 is also a CM local ring of dimension $n-1$. Using the inductive hypothesis, we conclude that $\underline{x}$ is an A-sequence. ■

THEOREM 5.6. Let $(A,\mathfrak{m})$ be a CM local ring.

(1) For a proper ideal I of A,

$$\operatorname{ht} I = \operatorname{depth}(I,A) \quad \text{and} \quad \operatorname{ht} I + \dim A/I = \dim A.$$

(2) A is catenary.

Proof. (1) Using Lemma 3.17, there is a sequence $\underline{a} = a_1,\dots,a_r \in I$ such that $\operatorname{ht}(a_1,\dots,a_r) = r$. In view of Theorem 5.5, $\underline{a}$ is an A-sequence, so that $\operatorname{depth}(I,A) \geqslant r$. On the other hand, if $\underline{b} = b_1,\dots,b_s \in I$ is an A-sequence, then due to Theorem 5.5, $\operatorname{ht}(b_1,\dots,b_s) = s \leqslant \operatorname{ht} I = r$, so that $\operatorname{depth}(I,A) \leqslant r$, whence $\operatorname{ht} I = \operatorname{depth}(I,A)$.

As for the second equality, let $\underline{a} = a_1,\dots,a_r \in I$ be an A-sequence with $r = \operatorname{depth}(I,A)$. Set $A' = A/(a_1,\dots,a_r)$ so that A' is a CM local ring of dimension $n-r$. Let I' denote the image of I in A'. Clearly $\operatorname{depth}(I',A') = 0$, so that $\operatorname{ht} I' = 0$ – in particular, I' is contained in one of the minimal primes of A. Further, note that $A/I \cong A'/I'$. For any minimal prime $\mathfrak{p}'$ in A', due to Theorem 5.4, $\dim A'/\mathfrak{p}' = \dim A'$, so that $\dim A'/I' = \dim A'$. It follows that

$$\dim A/I = \dim A'/I' = \dim A' = \dim A - r = \dim A - \operatorname{ht} I,$$

as desired.

(2) In view of Lemma 3.19, it suffices to show that for primes $\mathfrak{p} \subseteq \mathfrak{q}$, we have

$$\operatorname{ht} \mathfrak{q} = \operatorname{ht} \mathfrak{p} + \operatorname{ht}(\mathfrak{q}/\mathfrak{p}).$$

Due to Theorem 5.4, $A_{\mathfrak{q}}$ is a CM local ring, so that (1) gives

$$\operatorname{ht} \mathfrak{q} = \dim A_{\mathfrak{q}} = \operatorname{ht}\left(\mathfrak{p}A_{\mathfrak{q}}\right) + \dim \frac{A_{\mathfrak{q}}}{\mathfrak{p}A_{\mathfrak{q}}} = \operatorname{ht} \mathfrak{p} + \operatorname{ht}(\mathfrak{q}/\mathfrak{p}),$$

as desired. ■